Production Linkages and Regional Development: Evidence from Coal Mining in Germany*

Ann-Kristin Becker†

Job Market Paper.
Last update: 2025-11-17
Most recent version here.

Abstract

This paper examines how production linkages shape regional adjustment to industry shocks. I study this question by comparing hard coal and lignite mining in Germany. Both coal industries operated under the same institutional and macroeconomic conditions but differed markedly in their integration into local supply chains. Using a novel county-level employment dataset for Germany spanning 1849–2024, I estimate an event-study and a shift–share design. I find that hard coal mining, which produces strong upstream and downstream linkages, led to persistent increases in local manufacturing and service employment, particularly in input-supplying and coal-using industries. In contrast, lignite mining, with weaker linkages, led to local industrial decline, consistent with Dutch disease dynamics. I further show that the phase-out of hard coal caused substantial employment losses, particularly in formerly linked sectors. These results highlight the double-edged nature of production linkages: they fuel growth while a key industry is active, but can amplify local decline once it contracts.

Keywords: Production Linkages; Local Multiplier; Resource Extraction; Structural Change; Deindustrialization

JEL Codes: J21, O14, Q32, Q33, R11

*I am deeply grateful to Erik Hornung and Michael Krause for their invaluable support and guidance throughout this project. I would also like to particularly thank Guy Michaels for hosting me during a research visit at the LSE and for his generous feedback and encouragement, which significantly enriched this work. I further thank Rainer Fremdling, Richard Franke, Sebastian Braun, and Felix Kersting for sharing data. I also thank Björn Brey, Alan Fernihough, Andrew Goodman-Bacon, Tobias Korn, Ola Olsson, Carola Stapper, Jörn-Steffen Pischke, Nikolaus Wolf and the participants of the LSE Labour Economics Seminar 2024 (London, UK), the EHES 2025 (Hohenheim, Germany), the WEHC 2025 (Lund, Sweden), and seminar participants at Cologne for helpful comments and suggestions.

†University of Cologne, Albertus-Magnus-Platz, 50923 Cologne, Germany; (ann-kristin.becker@wiso.uni-koeln.de)

1 Introduction

Industry shocks produce divergent local outcomes, generating positive employment spillovers in some regions while causing stagnation or decline in others. Understanding how industry shocks affect regional economies is increasingly important. In many advanced economies, rising regional inequality has fueled political discontent, weakened social cohesion, and increased interest in industrial and place-based policy.[1]

Natural resource extraction is a particularly salient example. It has long shaped regional development, from the coalfields of Europe to the oil towns of North America and the mining zones of Latin America and Sub-Saharan Africa. Yet its local effects are strikingly uneven. Some mining regions experience sustained industrial growth, while others face deindustrialization or persistent decline.[2] Despite extensive research, the mechanisms behind this divergence remain poorly understood. Why do some resource-rich regions thrive while others fall behind?

In this paper, I examine one potential mechanism behind the heterogeneous local effects of natural resource extraction: the strength of local production linkages. Production linkages refer to the extent to which activity in one sector stimulates economic activity in other sectors through input demand and output supply (Hirschman, 1958). Isolating the role of production linkages is empirically challenging, as structural and institutional differences across regions and industries typically confound causal inference. To overcome this, I exploit a natural experiment in Germany (1849–2024). I compare the local effects of extracting two types of coal, hard coal and lignite. Both were mined under similar national institutions and at comparable scales, but differed markedly in their integration with the local economy. Hard coal extraction requires complex underground mining techniques and generates strong upstream linkages through demand for machinery, engineering, and metalworking. Its high energy density also made it a key input in a wide range of industries, creating substantial downstream linkages.[3] In contrast, lignite is extracted through simpler surface

[1] For industry shocks and local adjustment, see e.g. D. H. Autor et al. (2016), Black et al. (2005), Feyrer et al. (2017), Greenstone et al. (2010), and Hooker and Knetter (2001). On the political and social consequences of regional divergence, see e.g. D. Autor et al. (2019), D. Autor et al. (2020), Dippel et al. (2022), and Pierce and Schott (2020), Rodríguez-Pose (2018).

[2] A large empirical literature has documented both negative and positive long-run effects of resource extraction on local economic development. For negative effects, see e.g. Esposito and Abramson (2021), Papyrakis and Raveh (2014), and Shao et al. (2020); for positive effects, see e.g. Aragón and Rud (2013), Cavalcanti et al. (2019), and Michaels (2011). For overviews, see Havranek et al. (2016) and van der Ploeg (2011).

[3] *Upstream linkages* refer to input relationships where local industries supply goods or services to the mining sector. *Downstream linkages* refer to industries that use mining outputs (here, hard coal or lignite) as inputs in their

mining. In addition, due to its lower energy content, it was largely used for electricity generation and local heating. As a result, lignite mining is much less integrated into surrounding industries. This setting enables a within-country comparison of two closely related industries. Both coal industries operated within the same legal, institutional, and macroeconomic environment but differed markedly in their degree of local economic integration. This provides a credible framework to isolate the role of production linkages in shaping local economies, both during periods of industrial expansion and in the aftermath of sectoral decline.

To derive testable predictions about how extractive industries shape regional labor markets, I adapt the local multiplier framework from Moretti (2011). Specifically, I hypothesize that sectors like hard coal mining, which are embedded in local production networks, generate stronger spillovers to other tradable sectors and amplify demand for local services.[4] In contrast, sectors that operate as enclaves, such as lignite mining, are less likely to generate positive spillovers and may even crowd out other tradables by raising input costs.[5] Importantly, the very linkages that support growth during extraction may also increase exposure to decline. My framework implies that hard coal regions experience larger employment gains during the extraction period, but also sharp contractions during the coal phase-out. Lignite regions, by contrast, are likely to see few benefits, or even adverse effects, during extraction, but smaller losses during decline. I test these hypotheses empirically by analyzing the local effects of hard coal and lignite extraction in Germany during the coal extraction and phase-out period.

To examine whether differences in local linkage intensity shaped the long-term employment effects of resource extraction, I construct a novel county-level panel dataset for Germany from 1849 to 2024. I harmonize historical occupation titles from 28 censuses and align administrative boundaries, accounting for major political transitions from Prussia to modern-day Germany. My dataset spans more than 170 years and covers the rise, peak, and decline of both hard coal and lignite mining. To my knowledge, this is the first county-level dataset that captures local employment over such a long time span and industrial scope in Germany (Bartels et al., 2025; Berbée et al., 2025;

own production processes.

[4] This also echoes insights from staple theory, which emphasizes that the long-run impact of extracting a natural resource (the *staple*), depends on the extent to which it creates linkages to other sectors of the economy (Gunton, 2003; Watkins, 1963).

[5] Higher mining employment can raise local wages, increasing production costs for all firms. Since tradable goods are sold in national or international markets, local producers cannot pass these costs on to consumers and may become uncompetitive.

Peters, 2022). I link this dataset to historical input–output tables to classify manufacturing sectors by their upstream and downstream linkages to coal mining, enabling a disaggregated analysis of sectoral spillovers.[6]

To identify the local effects of coal extraction, I combine two complementary empirical strategies. First, I implement an event-study design comparing employment trends across hard coal, lignite, and non-coal regions after the onset of large-scale extraction and after the coal phase-out. To address concerns that mining activity may be endogenous to local conditions, I proxy for extraction using spatial variation in coal-bearing strata, which were formed long before modern economic activity and are plausibly exogenous (De Pleijt et al., 2020; Esposito & Abramson, 2021; Fernihough & O'Rourke, 2021).[7] Second, I estimate a shift–share specification that interacts annual variation in global coal prices with county-level exposure to coal deposits. This design exploits plausibly exogenous variation in exposure to global shocks to estimate the cyclical effects of extraction on local labor markets.

My findings show that the strength of local production linkages plays a central role in shaping the long-term effects of resource extraction. The event-study results show that the large-scale expansion of hard coal mining, characterized by strong local linkages, led to sustained gains in manufacturing employment. On average, manufacturing employment increased by approximately 19 percent relative to non-coal regions. These gains were driven by employment growth in linked industries, those supplying inputs to mining (upstream), those using coal as an input (downstream), and those with both types of linkages. Service employment also increased, consistent with local demand spillovers from rising income and population, while agricultural employment declined as regions industrialized. In contrast, lignite mining, marked by weaker local linkages, led to industrial decline. Manufacturing employment fell by approximately 20 percent alongside similar contractions in the service sector. These patterns suggest limited spillovers and crowding-out effects, consistent with Dutch disease dynamics (Corden & Neary, 1982; Krugman, 1987). Consistent with the sectoral responses, urban population increased in hard coal regions but not in lignite regions. This supports

[6] Upstream sectors include, for example, *Machinery* and *Wood products*, which supply equipment and materials to mining operations. Downstream sectors include *Primary metals* and *Nonmetallic mineral products*, which rely on coal as an energy input. Industries with weak linkages to coal include *Food and beverages*, *Apparel*, and *Printing and publishing*. The classification is based on the U.S. input–output table from 1947 (U.S. Bureau of Economic Analysis, 2019).

[7] I confirm that the distances from county centroids to hard coal and lignite fields are highly correlated with the distances from county centroids to carboniferous and lignite rock strata, respectively.

the interpretation that hard coal mining contributed to broader local development, whereas lignite mining did not.

I use the shift–share approach to complement the event-study findings by leveraging a distinct identification strategy that captures short-run responses to global coal price fluctuations. The results support the conclusion that coal booms raised labor demand in hard coal regions, but not in lignite areas. This reinforces the central role of production linkages in shaping the local effects of resource extraction.

After the phase-out of hard coal beginning in the 1960s, hard coal regions experienced pronounced deindustrialization. Manufacturing employment declined by about 21 percent relative to non-coal areas. A long-run event-study spanning 1849 to 2024 suggests that the employment gains in hard coal regions following the onset of large-scale extraction were largely offset by subsequent losses during the phase-out. The vulnerability of hard coal regions cannot be attributed to a lack of industrial diversification: hard coal regions became more diversified across manufacturing industries during the extraction period. Rather, my results point to the depth of integration with the coal economy as the key factor shaping the severity of employment losses following the decline of the hard coal industry. The decline of the lignite industry, which began in the 1990s, also had negative effects, but these were substantially smaller.[8]

Taken together, my results suggest that regions with a strongly integrated key industry tend to gain more during periods of industrial activity but face sharper contractions when that industry declines. In contrast, regions where a major industry is weakly linked to other local sectors tend to experience adverse effects during its operation, but more limited losses during its decline.

Contribution to the literature. This paper makes a central contribution to understanding the mechanisms through which industry shocks affect local economies. I compare two closely related coal industries, hard coal and lignite mining. Both operated under the same legal frameworks and were subject to the same national economic conditions. While they differed in extraction methods and industrial uses, the two sectors were otherwise very similar. This setting allows me to isolate the role of production linkages in a way that is difficult to achieve in cross-country or single-industry studies, where institutional and structural differences typically obscure the underlying mechanism.

[8]While lignite extraction in Germany has not yet been fully phased out, production has declined substantially since the 1990s.

Moreover, the long-run panel I construct allows me to examine how production linkages shape local labor market outcomes both during industrial activity and as it declines.

This paper mainly contributes to four strands of literature. First, it extends the literature on local multipliers (Greenstone et al., 2010; Moretti, 2011). I demonstrate that both the size and direction of industrial spillovers to tradable and non-tradable sectors can vary substantially, even within a single industry, depending on the intensity of local production linkages. This also informs debates on place-based and industrial policy by identifying the types of sectors most likely to generate large local multipliers (Austin et al., 2018; T. J. Bartik, 1991; Criscuolo et al., 2019; Juhász et al., 2024; Kline & Moretti, 2014; Lane, 2025; Moretti & Thulin, 2013; Neumark & Simpson, 2015; Sheremirov & Spirovska, 2022; Siegloch et al., 2025). My results suggest that policies targeting sectors with strong local linkages are more likely to generate durable local gains. In contrast, support for enclave-type industries with limited economic integration may even crowd out other sectors and constrain broader development.

Second, it contributes to research on structural change and regional resilience (Acemoglu et al., 2012; Barrot & Sauvagnat, 2016; Carvalho, 2014; Gagliardi et al., 2023). I show that even relatively diversified regional economies can become highly vulnerable when their lead sectors are tightly embedded in local input–output structures. Contractions in a locally embedded lead sector propagate to upstream suppliers and downstream users through production networks. These network effects amplify employment losses and trigger broader industrial decline. This propagation dynamic also helps explain slow or incomplete adjustment to large shocks documented in recent work (Berbée et al., 2025; Dix-Carneiro & Kovak, 2017; Esposito & Abramson, 2021; Franck & Galor, 2021; Fritzsche & Wolf, 2023; Glaeser et al., 2015; Jacobsen & Parker, 2016). My findings highlight the importance of forward-looking transition strategies. These are especially important under planned shifts such as decarbonization, where coordinated policies may mitigate negative network spillovers (Dechezleprêtre & Sato, 2017; Walker, 2013).

Third, this paper contributes to the literature on regional adjustment to industry shocks. I identify production linkages as a key mechanism shaping the long-term local effects of industrial expansion and decline. Prior work documents persistent local impacts of trade shocks, plant closures, and resource booms (D. H. Autor et al., 2016; A. W. Bartik et al., 2019; Black et al., 2005; Feyrer et al., 2017; Greenstone et al., 2010; Hooker & Knetter, 2001). But the channels through which these shocks propagate remain less understood. By comparing two closely related industries that differ in their local supply chain integration, I isolate the role of production linkages. My results show that they play a decisive role in determining how industry shocks affect local labor markets. These findings also speak to ongoing efforts to identify the relative importance of different sources of agglomeration economies (Ellison et al., 2010; Glaeser, 2008; Marshall, 1890; Rosenthal & Strange, 2001). By separating production linkages from other agglomeration channels, such as knowledge spillovers or labor pooling, this paper provides evidence on the importance of this mechanism in shaping local economic development.

Fourth, it adds to research on the local impacts of natural resource extraction (Allcott & Keniston, 2018; Aragón & Rud, 2013; Black et al., 2005; Lippert, 2014; Michaels, 2011; Pelzl & Poelhekke, 2021). I show that both the direction and magnitude of local effects depend on how deeply the extractive sector is integrated into the surrounding economy. This insight also informs the broader resource-curse debate (Cust & Poelhekke, 2015; Havranek et al., 2016; Sachs & Warner, 2001; van der Ploeg, 2011). My results show that variation in local linkage structures helps explain why resource extraction can lead to divergent long-run outcomes.[9]

Structure of the paper. The remainder of the paper is organized as follows. Section 2 provides background on Germany's coal mining industries, including evidence on differences in local supply chain integration between the hard coal and lignite sectors. Section 3 develops the conceptual framework and derives testable hypotheses based on Moretti (2010). Section 4 describes the data, and Section 5 outlines the empirical strategy and identification approach. Section 6 presents the main findings, focusing on the long-run economic impacts of coal mining, the cyclical effects of resource booms and busts, and the mechanisms driving the effects. Section 7 assesses the robustness of the results. Section 8 discusses the implications of my findings and concludes.

2 Setting: Hard Coal and Lignite Mining in Germany

Germany provides a compelling setting to examine how variation in production linkages shapes the long-run local effects of resource extraction. For over a century, the country extracted two major

[9] A related literature links resource booms to higher risks of violence and conflict, offering an additional channel for adverse local effects (Berman et al., 2017; Caselli & Michaels, 2013; Dal Bó & Dal Bó, 2011; Dube & Vargas, 2013).

coal types: hard coal (anthracite and bituminous) and lignite (brown coal). The two industries operated at comparable national scales and during overlapping periods. However, they differed substantially in their degree of integration into the surrounding economy. Hard coal was closely linked to upstream supplier industries and downstream industrial users, while lignite remained more economically isolated. This contrast enables a within-country comparison of similar resource sectors with different linkage intensity. It allows me to isolate the role of production linkages in shaping long-run regional development during both the expansion and decline of extractive activity.

2.1 Evolution of Coal Mining in Germany

While coal extraction in Germany began in the 18th century, it remained limited in scale until the mid-19th century. From that point onward, the diffusion of coal-based technologies from Britain both stimulated industrial demand and facilitated the expansion of mining activity (Tilly & Kopsidis, 2020).[10] Coal soon became the lead energy source for Germany's industrial economy.[11] By the early 20th century, Germany had become one of Europe's largest hard coal producers, with output levels comparable to Britain. Germany also emerged as the leading producer of lignite in Europe (Fremdling, 1995). While hard coal production initially dominated, lignite output caught up during the interwar period. From the 1920s onward, hard coal and lignite were produced at roughly similar scales (see Figure A1 in the Appendix).[12]

Germany's mining sector played a central role in the economies of coal-producing regions, both as a major employer and a land-intensive industry. At the beginning of the 20th century, more than 600,000 workers were employed across over 800 mines in the German Empire (Fischer, 2011). In 1907, mining accounted for over 8 percent of total employment in coal-producing counties.[13] In addition to its labor demands, coal mining required substantial land for extraction and infrastructure, making it both labor- and land-intensive.

[10] The number of steam engines in Prussia increased from 1,445 (29,482 horsepower) in 1849 to 35,431 (958,366 horsepower) in 1878, largely increasing demand for coal and facilitating its extraction (Banken, 1993; Tilly & Kopsidis, 2020).

[11] Between 1850 and 1900, Germany's hard coal production increased more than 20-fold, while lignite production rose more than 26-fold (see Figure A1a in the Appendix). Over the same period, employment in hard coal mining rose more than 12-fold, and lignite employment increased over 8-fold (own calculations based on data from Fischer (2011)).

[12] This catch-up was enabled by key technological innovations, most notably Exter's lignite briquetting press, which improved the energy density and transportability of lignite.

[13] Own calculations based on data from Kaiserliches Statistisches Amt (1910); see Section 4.2 for the definition of coal regions.

German hard coal production declined significantly by the 1960s. High labor costs, competition from cheaper imported coal, and the growing adoption of alternative energy sources, such as oil, natural gas, and nuclear power, undermined its competitiveness. A major factor was the influx of cheap oil from Northern Africa in the late 1950s, which triggered a sharp decline in labor demand in the mining sector (Fritzsche & Wolf, 2023). Between 1960 and 1985, hard coal production fell by nearly 50 percent, roughly 85 percent of mines closed, and employment in the sector declined by more than two-thirds.[14] The last hard coal mine in Germany closed in 2018. Lignite remained economically viable until the early 1990s as technological improvements in surface mining delayed its decline. By the 1990s, lignite output began to decline as stricter environmental regulations were introduced. Between 1990 and 2023, lignite output declined by more than two-thirds.[15] Although lignite extraction remains active today (as of 2025), it is scheduled to be fully phased out by 2038 as part of Germany's broader energy transition (KVBG, 2020). This long historical trajectory enables the analysis of both the short- and long-run effects of extractive activity and its subsequent decline.

2.2 Institutional Context

In Germany, hard coal and lignite were extracted under similar legal and administrative regimes. In the Prussian territories, including the Ruhr, Rhineland, and Lower Lusatia, the 1851 Co-ownership Act (*Miteigentümergesetz*) transferred mine management from the state to private actors, incentivizing investment, technological advancement, and large-scale coal extraction (Tilly & Kopsidis, 2020). This was followed by the 1865 General Mining Law (*Allgemeines Berggesetz*), which established uniform rules for mining concessions that applied equally to hard coal and lignite mining (ABG, 1865). Most other German states (e.g., Baden, Bavaria, Hesse, Württemberg) adopted similar legislation modeled on the Prussian framework (Deutscher Bundestag, 2024).[16] This legal alignment across coal types suggests that differences in long-run local outcomes between hard coal and lignite regions are unlikely to stem from variation in regulation or governance. Instead, they plausibly reflect differences in production linkages.

German coal extraction did not generate direct fiscal windfalls for municipalities through roy-

[14] Own calculations based on data from Fischer (2011).

[15] Own calculations based on data from Statista Research Department (2025a).

[16] A nationwide legal consolidation only occurred with the Federal Mining Act (*Bundesberggesetz*) in 1982 (BBergG, 1980).

alties. While local governments collected business tax revenue from local firms (*Gewerbesteuer*)[17] mining royalties accrued to the federal states (ABG, 1865; BBergG, 1980). This institutional arrangement limits the scope for direct fiscal linkages at the municipal level. Moreover, taxation rules did not differ between hard coal and lignite mining, suggesting that differential fiscal treatment is unlikely to explain observed variation in long-run outcomes across mining regions (Gunton, 2003; Watkins, 1963).

2.3 Geographic Distribution of German Coal Resources

Coal extraction in Germany is geographically concentrated in several distinct regions, with the two coal types—hard coal and lignite—mined in largely non-overlapping parts of the country. Major hard coal deposits are located in central and western Germany, particularly in the Ruhr region (e.g., the districts of Arnsberg and Düsseldorf), the Saar area (e.g., Trier), and in Upper Silesia (e.g., Oppeln, now part of modern Poland). Major lignite deposits are located in the Rhenish mining district (e.g., Köln), the Lusatian district (e.g., Frankfurt-Oder), and the Central German mining district (e.g., Merseburg) (Kunz, 2008). Figure 3 illustrates the distribution of coal fields in Germany, which closely corresponds to the historical location of coal mines (see also Section 4.2). The clear geographic separation between hard coal and lignite deposits, combined with their broad dispersion across Germany, provides the variation needed to credibly compare local economic outcomes across regions exposed to hard coal, lignite, or no coal mining, while holding institutions constant.[18]

2.4 Differences in Production Linkages Between Hard Coal and Lignite

Hard coal and lignite mining differed markedly in their potential to generate local economic spillovers through production linkages. I distinguish between two types of linkages: upstream linkages, reflecting the mining sector's demand for local intermediate inputs, and downstream linkages, reflecting its role as a supplier to other local industries.

[17] The *Gewerbesteuer* was introduced in 1891 (Preußisches Gewerbesteuergesetz, 1891).

[18] To address concerns about underlying structural differences between hard coal, lignite, and non-coal regions, I show in Section 7 that the three groups followed similar trends in key outcomes prior to large-scale coal extraction. In addition, I demonstrate that the results are robust to the inclusion of district-by-year fixed effects, which absorb time-varying shocks at the subnational level.

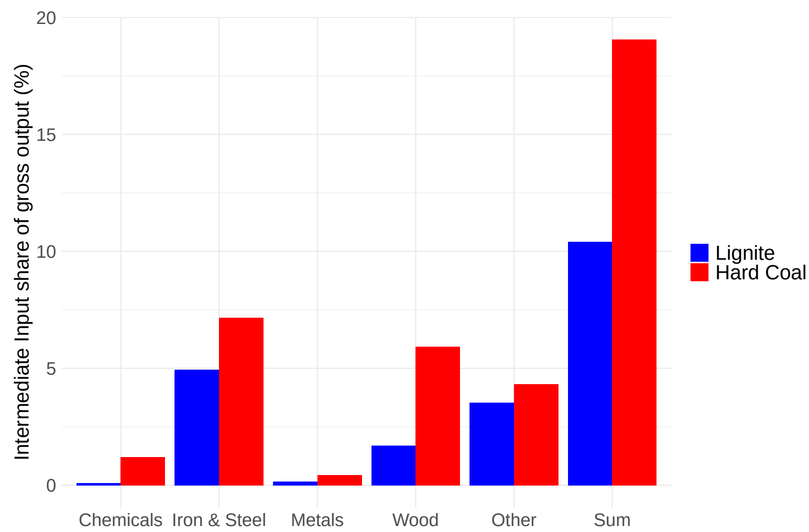


Figure 1: Hard Coal and Lignite Mining Intermediate Inputs in Germany (1936)

Notes: The figure plots the value of inputs purchased from various sectors by the lignite (blue) and hard coal (red) mining industries in Germany in 1936, expressed as a share of each sector's total output. Own calculations based on data from the Statistisches Reichsamt (1936a).

2.4.1 Upstream Linkages: Coal Mining's Dependence on Local Inputs

Hard coal mining relies more heavily on upstream industries than lignite mining, reflecting its more complex extraction process. Unlike lignite, which is typically found near the surface and extracted via open-pit methods, hard coal lies deep underground. Extracting it requires shafts, tunnels, drainage systems, and specialized equipment. This more demanding extraction process creates greater demand for machinery, metal components, industrial materials, and related services (Hovis & Mouat, 1996; Pierenkemper, 1992).[19]

Figure 1 illustrates this stark difference in input intensity. In 1936, intermediate goods accounted for 19.1% of total production value in hard coal mining, compared to just 10.4% in lignite. Hard coal also required a higher share of specialized materials, such as mine timber and explosives, which made up 6.3% of hard coal output versus only 1.5% for lignite (Statistisches Reichsamt (1936a), own calculations).

Where economic and institutional conditions permit, mining firms often source inputs locally to reduce transport costs, improve reliability, and allow for customized supply (Ellison & Glaeser,

[19]See Figure A4 in the Appendix for photographs illustrating the extraction methods of lignite and hard coal mining.

1997; Moretti, 2011; Morris et al., 2012). Consequently, the greater input intensity of hard coal extraction results in stronger upstream linkages to local suppliers compared to lignite.

2.4.2 Downstream Linkages: Local Use of Coal

Hard coal supports a broader range of industrial applications than lignite, particularly in energy-intensive sectors, resulting in stronger downstream linkages. By contrast, lignite's lower energy density, higher moisture content, and lower carbon concentration limits its industrial use.[20]

The differences in energy content and usability between hard coal and lignite translates into distinct patterns of industrial coal use, as shown in Figure 2. Lignite consumption was more heavily concentrated in electricity generation, with relatively smaller use in industrial production compared to hard coal. Hard coal, by contrast, was primarily used in industrial processes, particularly in heavy manufacturing, metal processing, and transport.

Processing coal near extraction sites offered important economic advantages. Coal is bulky, degrades quickly, and is expensive to transport and store. These constraints encourage spatial co-location between coal extraction and industrial production, particularly in hard coal regions (Ellison & Glaeser, 1997). As a result, downstream spillovers through industrial demand are far more pronounced in hard coal regions than in lignite regions.

Taken together, the stark difference in linkage intensity between the hard coal and lignite industries, despite similar institutional contexts, provides a rare opportunity to isolate how input–output structures shape the long-run regional effects of resource extraction.

[20]Lignite's chemical composition also prevents its use in coke production, a crucial input for blast furnaces and heavy industry.

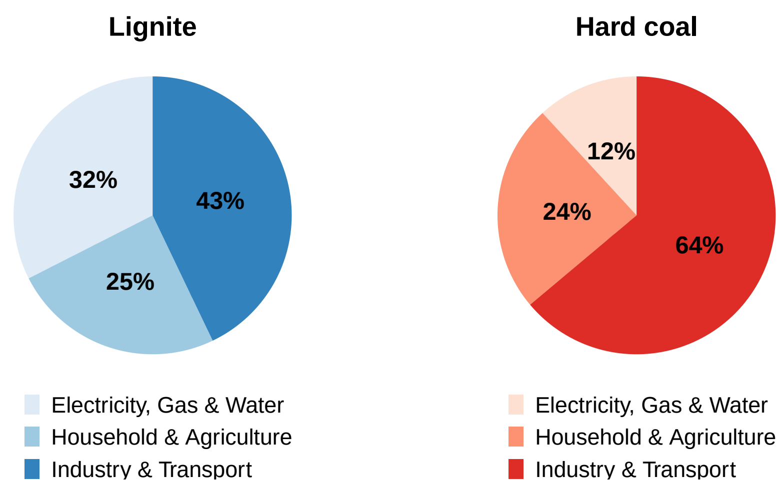


Figure 2: Hard Coal and Lignite Use in Germany (1934)

Notes: The figure shows the sectoral shares of lignite (left) and hard coal (right) use in Germany in 1934. Own calculations based on data from the Statistisches Reichsamt (1934).

3 Conceptual Framework: Sectoral Spillovers from Coal Mining

To study how coal mining affects employment across sectors depending on linkage intensity, I develop a simple general equilibrium framework adapted from Moretti (2011). I model each county as a competitive economy that produces tradable goods (mining, manufacturing, agriculture) sold on national markets and non-tradable goods (services) whose prices are set locally.[21] Labor is homogeneous and mobile across sectors within a county, implying that wages equalize locally across industries. The local labor supply curve is upward sloping: workers' utility depends on local wages net of living costs, along with idiosyncratic location preferences. Housing supply is also upward sloping, reflecting geographic constraints and land-use regulation. Amenities are assumed to be constant across space. This framework allows me to derive testable predictions about how extractive sectors with varying linkage intensities shape local labor markets.

Spillovers to the Tradable Sector. An increase in mining employment raises local wages and prices, which in turn increases production costs for all local firms. Since tradable goods are sold in national or international markets, local producers cannot pass these higher costs on to consumers (they are price-takers), potentially leading to crowding-out effects. However, if mining generates upstream or downstream linkages with other local industries, it may stimulate activity

[21] This distinction follows Moretti (2011), based on the geographic scope of product markets.

in related sectors. The net impact depends on the strength of these production linkages and other agglomeration forces, such as knowledge spillovers or labor pooling, relative to the cost effect.

In the case of hard coal, both upstream and downstream linkages are strong, as outlined in Section 2.4. Mining depends on industrial inputs and simultaneously supplies fuel to a broad range of industries. These linkages are likely to generate positive spillovers to nearby manufacturing firms, which may expand employment in response to increased input demand from mining or improved access to coal as an input. By contrast, agriculture, lacking direct connections to mining, primarily faces cost-side pressures: rising wages and land prices reduce profitability, potentially leading to employment declines.

> **Hypothesis H1:** *Manufacturing employment increases in hard coal regions, particularly in sectors with strong upstream or downstream linkages to coal mining.*
>
> **Hypothesis H2:** *Agricultural employment declines in hard coal regions due to rising input costs.*

In lignite regions, production linkages are relatively weak. Lignite is primarily used for local heating and electricity generation, and its extraction process relies on fewer intermediate inputs. As a result, manufacturing firms benefit little from proximity to mining activity. However, they still face upward wage pressure, which can reduce manufacturing employment. The effects on agriculture are more ambiguous. On the one hand, rising wages may increase production costs; on the other hand, if mining displaces other sectors, wage pressures may ease over time, potentially stabilizing agricultural employment.

> **Hypothesis L1:** *Manufacturing employment declines in lignite regions due to weak linkages and rising input costs.*
>
> **Hypothesis L2:** *Agricultural employment remains stable or increases in lignite regions if wage pressures are offset by reduced demand in other sectors.*

Spillovers to the Non-Tradable Sector. Higher mining wages and employment raise local incomes, which in turn increase demand for non-tradable goods and services. The magnitude of this demand linkage effect depends on: (1) the skill and earnings profile of mining-sector jobs; (2) consumer preferences for non-tradables, i.e., the share of additional income spent locally; (3) the labor intensity of service industries; and (4) general-equilibrium wage and price adjustments, which are shaped by the elasticities of local labor and housing supply. In addition, mining operations may generate direct service demand for activities such as logistics, maintenance, and catering.

In hard coal regions, both income-driven and mining-related service demand are expected to raise employment in the non-tradable sector. In lignite regions, by contrast, weaker income gains and limited service linkages may result in stagnant or declining service employment, particularly if population falls due to out-migration or crowding-out of other sectors.

> **Hypothesis H3:** *Service employment increases in hard coal regions due to higher income and mining-related demand.*
>
> **Hypothesis L3:** *Service employment stagnates or falls in lignite regions due to limited income growth and weak service demand from mining.*

Total Employment Effects. In regions with strong input–output linkages, mining acts as a catalyst for broader economic expansion. Hard coal regions should experience net employment growth across non-mining sectors, despite agricultural losses. In lignite regions, weak spillovers and higher costs may lead to net employment losses in non-mining sectors.

> **Hypothesis H4:** *Total non-mining employment increases in hard coal regions due to strong local multipliers.*
>
> **Hypothesis L4:** *Total non-mining employment declines in lignite regions due to weak spillovers and crowding-out effects.*

Effects of Coal Phase-Outs. The framework also applies to gradual declines in mining activity, such as those associated with long-term coal phase-outs. These contractions reverse the mechanisms described above.

In hard coal regions, strong upstream and downstream linkages imply that declining mining employment reduces demand in manufacturing and services, triggering broader employment losses. In lignite regions, by contrast, impacts are more concentrated within mining itself. Still, falling incomes and population may reduce demand for local services, leading to additional employment losses in the non-tradable sector.

Hypothesis H5: *A decline in hard coal mining reduces employment in manufacturing and services due to strong production and demand linkages. Effects on agriculture are ambiguous and may depend on how wage and land cost pressures adjust.*

Hypothesis L5: *In lignite regions, employment losses are concentrated in services, while manufacturing and agriculture are less affected.*

I test these predictions using two complementary empirical strategies. First, I employ an event-study framework to examine the long-run effects of mining and the coal phase-out. Second, I implement a shift–share design that interacts global coal price shocks with county-level exposure to coal endowments, capturing the dynamic effects of resource booms. Together, these strategies allow me to isolate the causal impact of coal extraction and distinguish between short-run boom–bust cycles and persistent long-term effects.

4 Main Data

This section describes the construction of the main dataset used in the analysis. The unit of observation is the county. To ensure spatial consistency over time, I harmonize administrative boundaries to match those in place in 1900 (see Section A.3 in the Appendix).

4.1 Outcome Variables: County-Level Sectoral Employment 1849–2024

To analyze the local effects of coal mining, I compile a novel county-level employment dataset for Germany spanning 1849 to 2024. The dataset captures the rise and expansion of the hard coal and lignite industries as well as the onset of the hard coal phase-out in the 1960s and the decline of lignite mining starting in the 1990s. To construct a consistent employment panel, I harmonize sectoral employment classifications across 28 censuses. For most years, employment is observed at the 2- or 3-digit sector level, enabling disaggregated analysis of sectoral spillovers. To ensure spatial consistency over time, I align administrative boundaries from Prussia to modern-day Germany to ensure spatial comparability (see Section A.3). To my knowledge, this is the first dataset to track county-level employment in Germany over a period of more than 170 years (Bartels et al., 2025; Berbée et al., 2025; Peters, 2022).[22] This long-run panel allows me to trace the economic effects of

[22] Descriptive statistics are reported in Table A2 in the Appendix. Section A.4 describes the underlying sources.

coal mining during both the extraction period and the subsequent decline.

Disaggregated Manufacturing Employment and Production Linkages. To examine heterogeneity in the effects of coal mining across manufacturing industries, I construct measures of upstream and downstream linkages between each industry and the mining sector. These linkage intensities are derived from the 1947 U.S. Input-Output (I-O) table compiled by the U.S. Bureau of Economic Analysis (2019), which provides detailed information on inter-industry transaction flows.[23]

For each manufacturing industry, I calculate first- and second-degree upstream and downstream linkage shares. The first-degree downstream linkage is defined as the share of an industry's total inputs sourced directly from the mining sector. The mining sector itself is excluded from linkage calculations:

$$\text{FirstDegreeDownstream}_i = \frac{\text{InputFromMining}_i}{\text{TotalInput}_i}$$

The first-degree upstream linkage is defined as the share of an industry's output sold directly to the mining sector:

$$\text{FirstDegreeUpstream}_i = \frac{\text{OutputToMining}_i}{\text{TotalOutput}_i}$$

Second-degree linkages capture indirect relationships via intermediate industries and are calculated analogously through two-step I-O matrix multiplications. An industry is classified as upstream-linked if its total upstream linkage (direct plus indirect) exceeds the sector median. To prevent electricity-related sectors from dominating the downstream classification, I follow Allcott and Keniston (2018) and use only the direct input share.[24] Industries are considered downstream-linked if their direct input share from mining exceeds the median. An industry is defined as linked to mining, if it is classified as either upstream- or downstream-linked; otherwise, it is considered non-linked.

Appendix Table A4 reports the linkage shares and binary indicators for each manufacturing industry. The strongest upstream linkages to coal mining are observed in *Machinery* and *Wood*

[23] I rely on the U.S. I-O table to avoid potential endogeneity concerns. Historical German I-O tables may themselves have been shaped by coal mining activity. In robustness checks, I confirm that the results are not sensitive to the choice of I-O table.

[24] As shown in Section 7, results are similar when using either direct or indirect input shares for both upstream and downstream linkages.

products, both of which supply key inputs to mining operations. Industries with the highest downstream linkages to coal mining are *Nonmetallic mineral products* and *Primary metals*, reflecting their reliance on coal as an energy input. By contrast, industries with weaker input–output linkages to coal include *Food and beverage and tobacco products*, *Apparel and leather and allied products*, and *Printing and publishing*. Finally, I map census occupations for each available year to the I-O sectors, allowing disaggregated analysis of sector-specific employment dynamics.[25]

Sectoral Diversification of Manufacturing Employment. To assess whether coal mining influenced the structure of local economies, I examine its effect on industrial diversification across regions. I measure diversification using a Herfindahl–Hirschman Index (HHI) of manufacturing employment. I calculate the HHI at the county-year level for each census year between 1849 and 1939 in which disaggregated occupational data are available. The index is defined as the sum of squared employment shares across manufacturing industries:

$$\text{HHI}_{ct} = \sum_{i \in I} \left(\frac{\text{Employment}_{ict}}{\text{TotalEmployment}_{ct}} \right)^2,$$

where $i \in I$ indexes manufacturing industries observed in county c and year t. Higher index values reflect greater concentration, while lower values indicate a more diversified manufacturing structure. This measure allows me to examine whether coal extraction led to increased industrial specialization or diversification in coal-endowed regions. These structural patterns may have shaped regional vulnerability and influenced how areas responded to the subsequent coal phase-out.

4.2 Treatment Variables: Hard Coal and Lignite Endowments

To measure exposure to coal mining, I use geological data on the location of hard coal and lignite rock strata across Germany. Rather than relying on historical mining activity, which may be endogenous to prior economic development, I follow A.-K. Becker and Hornung (2025), De Pleijt et al. (2020), Esposito and Abramson (2021), and Fernihough and O'Rourke (2021) in using geological resource endowments, based on data from Asch and Bellenberg (2005), as a plausibly exogenous

[25] While I observe total manufacturing employment data for 28 census years between 1849 and 2024, disaggregated industry-level data (at the 2- or 3-digit level) are only available for the years 1849, 1882, 1895, 1907, 1939, and from 2007 to 2024.

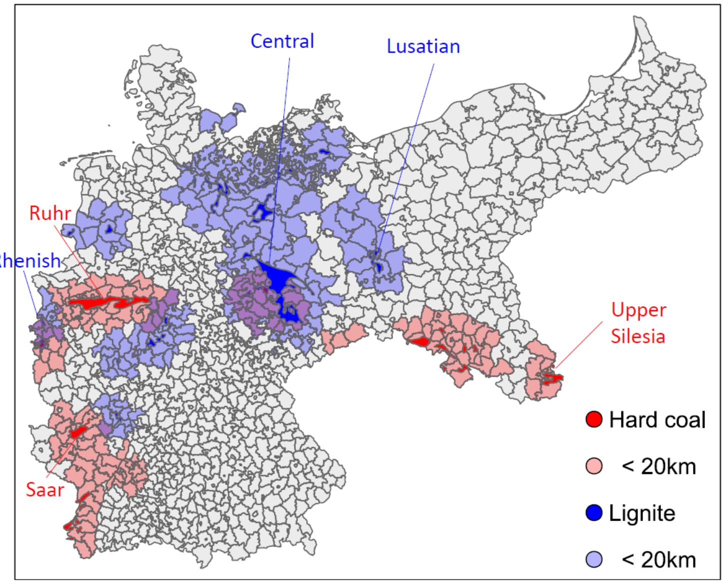


Figure 3: Geographic Locations of Hard Coal and Lignite Deposits in Germany

Notes: The map displays hard coal (red) and lignite (blue) rock strata within the borders of Germany as of 1900. Counties classified as hard coal or lignite regions in the analysis, i.e. those located within 20 kilometer (km) of the respective deposits, are highlighted accordingly. Major coal-producing areas (e.g., Ruhr, Saar, Upper Silesia) and lignite basins (e.g., Rhenish, Central German, Lusatian) are labeled. Data source: Asch and Bellenberg (2005).

proxy for mining potential.

I classify strata from the Carboniferous period (ca. 320–295 million years ago) as hard coal-bearing, since nearly all German hard coal was formed during this era. Lignite, by contrast, is typically found in younger formations dating from the Miocene to Pliocene periods (ca. 60–5 million years ago). Because these younger formations are geologically more heterogeneous, I restrict lignite classification to areas where lignite is among the two most prevalent rock types. I then compute the distance from each county's geographic centroid to the nearest hard coal or lignite-bearing strata. As shown in Figure A2a and Figure A2b in the Appendix, the distances from county centroids to hard coal and lignite fields are highly correlated with the distances from county centroids to carboniferous and lignite rock strata, respectively. I classify a county as a hard coal (or lignite) region if its centroid lies within 20 km of the corresponding strata. This threshold reflects a historically plausible commuting distance. Alternative thresholds are tested in robustness checks (see Section 7).

Table A1 in the Appendix presents summary statistics. On average, counties are located 132 km from the nearest hard coal deposit and 110 km from the nearest lignite deposit. Based on the

20 km cutoff, 16% of counties are classified as hard coal regions and 18% as lignite regions. About 30% of counties are located near either type of deposit, with roughly 3% lying near both.

4.3 Geographic Controls

I account for several geographic factors that may confound the relationship between coal endowments and local labor market outcomes. First, to account for differences in market access, I control for the distance to the nearest navigable river and the coast, both of which may influence employment patterns and may be correlated with coalfield locations. Second, I include climatic and agricultural suitability measures, average temperature, precipitation, and a caloric suitability index from Galor and Özak (2016), to capture variation in agricultural productivity that may affect sectoral employment. Third, I control for terrain slope to account for the historical use of water mills, the main alternative to steam engines prior to widespread coal adoption (Gutberlet, 2014). Steeper terrain was more suitable for water-powered mills, which facilitated early proto-industrial development. As a result, these regions may have accumulated more physical capital and been better positioned to adopt new technologies, with potential long-run effects on employment patterns (Ashraf et al., 2025).[26]

5 Empirical strategy

5.1 Event-study: Long-Run Effects of Resource Extraction and Its Phase-Out

To assess the long-run effects of hard coal and lignite mining on local economies and to evaluate the effects of the coal phase-out, I estimate the following event-study specification:

$$Y_{ct} = \sum_{j=t_1}^{T} \beta_j (HardCoal_c \times I_t^j) + \sum_{j=t_1}^{T} \delta_j (Lignite_c \times I_t^j) + \sum_{j=t_1}^{T} (X_c' I_t^j)\kappa_j + \alpha_c + \gamma_t + \varepsilon_{ct}. \qquad (1)$$

Here, Y_{ct} is the outcome of interest in county c and year t.[27] The indicators $HardCoal_c$ and $Lignite_c$ equal 1 if the county centroid lies within 20 km of a hard coal or lignite deposit, respectively. These indicators are interacted with year dummies I_t^j to estimate time-varying treatment effects.

[26] Table A1 in the Appendix provides descriptive statistics.
[27] t is observed in years between 1849 and 2024 for which census data are available: 1849, 1882, 1895, 1907, 1939, 1950, 1961, 1970, 1987, and annually from 2007 to 2024.

The coefficients β_j and δ_j represent the differential outcomes for hard coal and lignite counties, respectively, compared to non-coal counties in year j, relative to the baseline year. Due to changes in territorial boundaries, I estimate the model separately for (i) the period of coal industry expansion and (ii) the era of rapid coal decline, using distinct time windows and baseline years for each subsample. For the extraction period, I restrict the sample to Prussian counties. For the decline period, I restrict the sample to West German counties. I also estimate the model over the full 1849–2024 period to assess net long-run effects. For this specification, the sample is restricted to counties in the intersection of Prussia and modern-day Germany.

The vector X_c includes time-invariant geographic controls (temperature, rainfall, slope, crop suitability, distance to the nearest river and coast) interacted with year fixed effects to flexibly account for time-varying confounders. County fixed effects α_c absorb all time-invariant unobserved heterogeneity, while year fixed effects γ_t control for common shocks affecting all counties. Standard errors are clustered at the county level.

Identification relies on the standard parallel trends assumption.[28] Let treatment be the positive labor demand shock associated with the rapid expansion of industrial-scale coal mining in Germany beginning in the mid-19th century. In the absence of this event, treated counties (with coal deposits) and control counties (without deposits) would have followed similar trends in the outcome, conditional on controls and fixed effects. This assumption is plausible, as coal deposits were determined by natural processes long before the onset of modern economic activity. I further assess the validity of this assumption by examining pre-trends in alternative outcomes (wages and urban population) in Section 7. For the decline period, treatment captures the negative labor demand shock associated with the contraction of the coal sector, primarily driven by the influx of cheap oil starting in the late 1950s. In this case as well, identification rests on the assumption that, absent the decline, treated and comparison counties would have exhibited parallel trends. I assess this assumption by reporting pre-trends in the main outcomes.

[28] Because treatment is defined as a time-invariant exposure (historical coal deposits), issues related to staggered adoption timing in two-way fixed effects models do not apply (Callaway & Sant'Anna, 2021; Goodman-Bacon, 2021; Sun & Abraham, 2021). Moreover, the estimates should be interpreted as reduced-form, intention-to-treat style effects of exposure, rather than effects of realized mine openings, closures or mining intensity.

5.2 Reduced form Shift-Share Approach: Effects of Coal Booms

To complement the event-study analysis, I estimate a reduced-form "shift-share" design that exploits plausibly exogenous variation in exposure to global coal market fluctuations. Exposure is measured by proximity to geological coal deposits, while global coal prices provide the time-varying shocks.

$$\begin{aligned} Y_{ct} = {} & \beta_H \, \text{HardCoal}_c \times \text{WorldPrice}_t \\ & + \beta_L \, \text{Lignite}_c \times \text{WorldPrice}_t + \sum_{j=t_1}^{T} X_c' I_t^j \kappa_j + \alpha_c + \gamma_t + \epsilon_{ct}. \end{aligned} \tag{2}$$

Here, $HardCoal_c$ and $Lignite_c$ are binary indicators equal to 1 if county c lies within 20 km of a hard coal or lignite deposit, respectively. The variable $WorldPrice_t$ is the (log) annual average world coal price, deflated to constant prices and standardized to mean zero and unit standard deviation (sd).[29] It captures the contemporaneous[30] price shock in year t.[31] Coefficients β_H and β_L capture the differential effect of a one sd increase in global prices on outcomes in hard coal and lignite regions, respectively, relative to non-endowed counties. All specifications include county fixed effects (α_c), year fixed effects (γ_t), and a vector of geographic controls interacted with year fixed effects ($X_c' I_t^j$).

This design complements the event-study framework by leveraging a distinct source of identifying variation. While the event-study relies on parallel trends, identification here requires that, conditional on controls, there is no unobserved factor that is both correlated with a county's geological coal endowment and systematically co-moves with global coal prices. The exposure measure is plausibly exogenous because the location of coal deposits is time-invariant and determined by ancient geological processes unrelated to modern economic activity (Goldsmith-Pinkham et al., 2020). The shift component, movements in the world coal price, is likewise plausibly exogenous to local German conditions (Adao et al., 2019; Borusyak et al., 2022). To further mitigate con-

[29] Separate world price series for hard coal and lignite are not available. The available aggregate world coal price largely reflects hard coal price movements, as lignite was rarely traded internationally. However, German hard coal and lignite prices display strong year-to-year co-movement during 1882–1939 (correlation of annual changes $\rho = 0.87$, $p < 0.001$), indicating exposure to common cyclical shocks. This supports the use of a single global coal price as a proxy for both fuel types. As a robustness check, I use national-level series that allow for coal-type-specific shocks (see Section 7).

[30] To account for potential delayed responses, I use a five-year moving average of the price shock in a robustness check (see Section 7).

[31] Price data are available for all years from 1882 to 1939.

cerns, I include county and year fixed effects, a rich set of geographic controls interacted with year dummies, and robustness checks using placebo shocks and alternative shift variables to support the identifying assumption (see Section 7).

The shift–share design also offers several practical advantages. By leveraging exogenous variation from global coal booms and busts, I begin the analysis of the extraction period in 1882. This enables inclusion of all counties within the historical borders of the German Empire. In contrast, the event-study analysis for the coal extraction period is restricted to Prussian counties. Additionally, price shocks provide more precise timing of economic cycles than the gradual and continuous treatment used in the event-study. While local labor market responses may not be immediate, due to adjustment frictions, the shift–share design helps distinguish short-run boom–bust dynamics from longer-run structural effects of resource extraction and reinforces identification through an alternative source of variation.

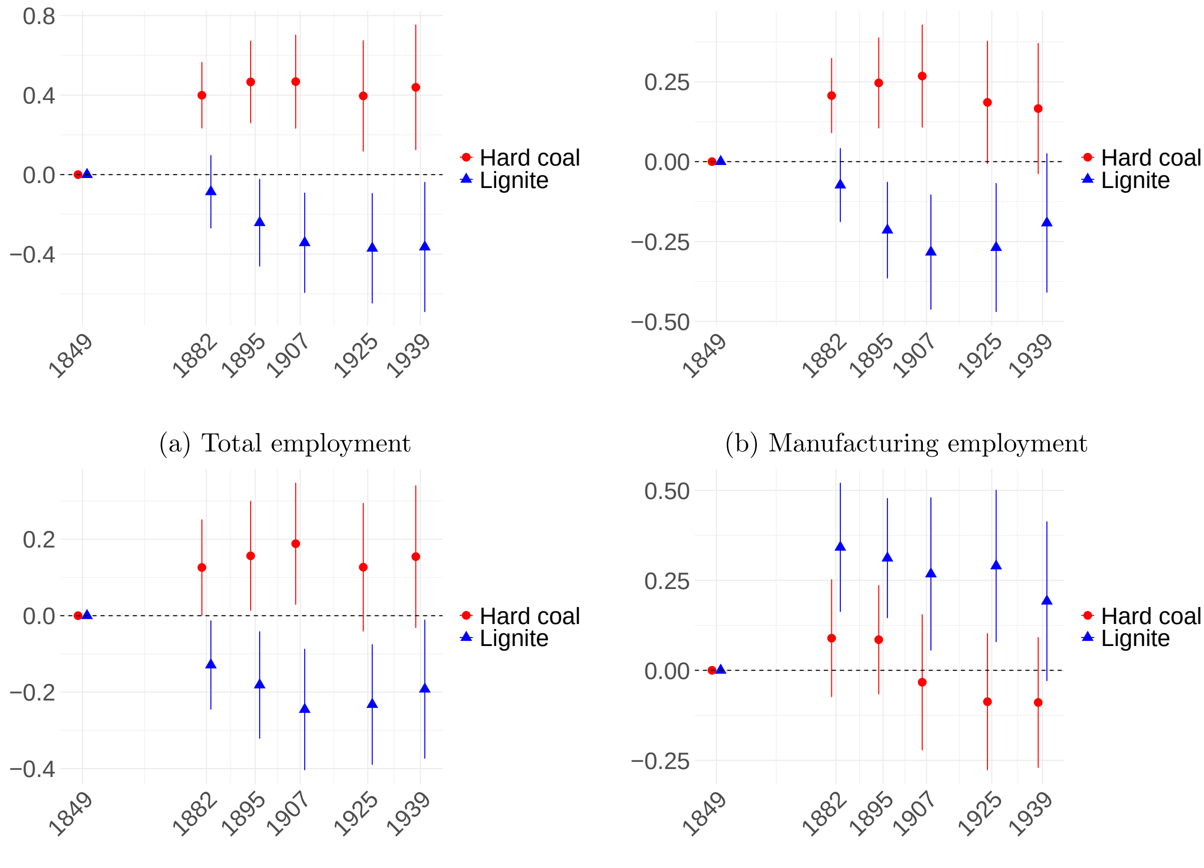
6 Main Results

6.1 Long-Run Effects of Resource Extraction: Event-study Results

I begin by examining the long-run labor market effects of the positive labor demand shock generated by the onset of large-scale coal extraction. The year 1849 serves as the pre-treatment baseline, as coal production was limited to small-scale operations prior to this point. Unfortunately, no occupational censuses were conducted before 1949. To assess the robustness of the parallel trends assumption, I examine trends in alternative proxies for regional development and growth potential in Section 7. Industrial-scale mining expanded rapidly in the decades after 1849, driven by advances in extraction technologies and rising industrial demand from the adoption of coal-using technologies. I therefore expect potential spillover effects to emerge by the first post-baseline year, 1882 (see Figure A1 in the Appendix).[32] The event-study estimates thus capture employment changes after the onset of large-scale coal extraction.

Figure 4 shows that the local effects of coal extraction varied sharply by coal type, consistent with the role of production linkages outlined in Section 3. I begin by examining the effects of coal

[32] The analysis is restricted to counties that belonged to Prussia in 1849, as occupational census data for that year are only available for this subset. I end the analysis in 1939 to avoid confounding effects from the major border changes that occurred after World War II.



(a) Total employment

(b) Manufacturing employment

(c) Service employment

(d) Agricultural employment

Figure 4: Event-Study Estimates: Effect of Resource Endowment on Sectoral Employment (1849–1939)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties. Corresponding estimates for each plot are presented in Table A6 in the Appendix.

extraction on total employment, defined as the sum of agricultural, manufacturing, and service sector employment. Mining employment is excluded, as data for this sector are not consistently reported across all benchmark years and are unavailable for the baseline year (1849). As a result, the estimates for total employment capture only indirect employment effects and understate the full employment impact of coal extraction. Total employment (excluding mining) increased by 0.4 to 0.5 (log) sd between 1849 and 1882 in hard coal regions compared to non-coal regions after 1849

(Panel 4a). Employment remained persistently elevated through multiple boom-and-bust periods until 1939.[33] In contrast, lignite regions saw a sustained decline in total non-mining employment between 1882 and 1939 below baseline, consistent with limited mining spillovers and crowding-out of other sectors. These patterns align with Hypothesis H4, which anticipates broad employment gains in hard coal regions following large-scale mining expansion, driven by strong production linkages and local multipliers. They are also consistent with Hypothesis L4, which predicts limited or negative employment effects of lignite extraction due to weak intersectoral connections.[34]

In manufacturing (Panel 4b), employment in hard coal regions rose by 0.2 to 0.3 (log) sd, in line with strong upstream and downstream linkages (Hypothesis H1). Lignite regions saw declines of similar magnitude, consistent with weak production linkages and crowding-out effects (Hypothesis L1). A similar pattern emerges in the service sector (Panel 4c): employment in hard coal regions rose, likely reflecting demand spillovers from rising population, income, and mining-related service needs (Hypothesis H3). By contrast, service employment in lignite regions declined, likely due to weaker local demand and limited demand-side linkages from mining activity (Hypothesis L3). In agriculture (Panel 4d), hard coal regions show no significant change, consistent with offsetting forces from wage pressures (Hypothesis H2). In contrast, agricultural employment in lignite regions increases modestly, likely reflecting reallocation into lower-productivity sectors as manufacturing and services contracted (Hypothesis L2).

To assess the magnitude of the employment effects more transparently, Table A7 in the Appendix reports estimates from a difference-in-differences (DiD) specification that replaces the event-time indicators in Equation 1 with a post-1849 indicator. Relative to 1849, hard coal regions experienced increases of 0.42 (log) sd in total non-mining employment (approximately 25% in levels), 0.21 (log) sd in manufacturing (19%), and 0.14 (log) sd in services (16%). By contrast, lignite regions saw a decline of 0.22 (log) sd in manufacturing employment, equivalent to roughly 20% in levels.

[33] Despite an overall rise in hard coal output between 1882 and 1939, the employment gap between hard coal and non-coal regions does not widen further. This pattern is likely explained by capital deepening and productivity gains in both mining and coal-linked industries, which reduced labor demand per unit of output.

[34] Moretti and Thulin (2013) and Moretti (2010) note that local multipliers for tradable industries represent lower bounds for national multipliers. Because the market for tradables is national, much of the additional demand generated by local shocks, such as coal expansion, also benefits other regions. This implies that the positive employment spillovers observed in hard coal regions likely understate the aggregate national gains from hard coal–related linkages. At the same time, negative cost effects, such as higher local wages, tend to be more pronounced at the local level than nationally. Consequently, the negative employment effects in lignite regions, reflecting weak production linkages and predominantly higher input costs, are likely less pronounced at the national level.

To directly assess heterogeneity within the coal sector, I restrict the sample to coal-endowed counties. This specification compares manufacturing employment in hard coal versus lignite regions, holding constant the presence of mining activity. The DiD result implies a 0.26 (log) sd increase (22%) in manufacturing for hard coal relative to lignite counties following the onset of large-scale coal extraction (see Column (6) of Table A14 in the Appendix). This within-coal comparison reinforces the interpretation that variation in local production linkages drives the divergent regional outcomes.

Consistent with the sectoral employment responses, Figure A15 in the Appendix shows that hard coal regions experienced substantial and sustained increases in urban population following the onset of large-scale mining. Urban population in these regions grew steadily from the mid-19th century onward, with gains persisting through 1939. In contrast, lignite regions exhibited no urban population growth over the entire period. Importantly, there are no differential trends in urban population across hard coal, lignite, and non-coal regions prior to the onset of large-scale mining (1816–1849), supporting the identification assumption (see also Section 7). These patterns reinforce the interpretation that hard coal mining contributed to broader local development, while lignite mining did not.

These findings highlight how the strength of local production linkages shaped the long-run regional effects of extraction. In hard coal areas, strong integration into industrial supply chains supported broad-based growth, that persisted across multiple boom and bust cycles. In lignite regions, by contrast, weak linkages limited spillovers and led to deindustrialization. Given the lower long-term productivity potential of extractive sectors, this likely constrained regional growth (Jacobsen & Parker, 2016; van der Ploeg, 2011).

6.2 Effects of Resource Booms: Results of the Reduced-Form Shift-Share Approach

I now turn to the estimates from the reduced-form shift–share approach from Equation 2, which complement the event-study results by exploiting time-series variation in global coal prices interacted with local geological coal endowments.[35] This approach allows me to estimate how

[35] Figure A3 in the Appendix plots annual global coal prices, illustrating boom and bust periods. Descriptive statistics are reported in Table A3 in the Appendix.

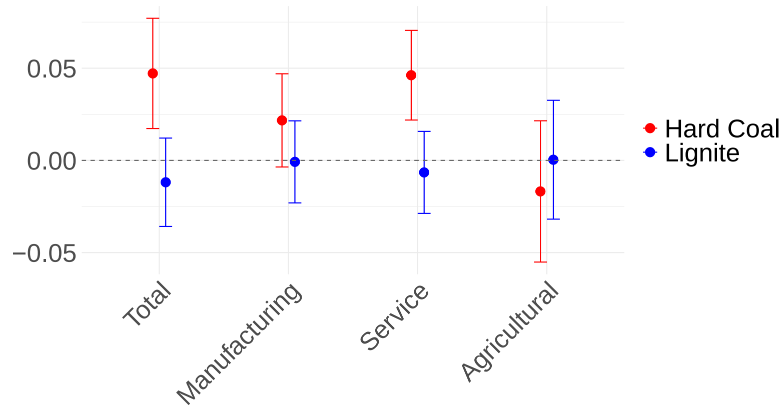


Figure 5: Reduced-Form Shift-Share Estimates: Effect of Resource Booms on Sectoral Employment (1882–1939)

Note: This figure plots point estimates and 95% confidence intervals from reduced-form shift-share approach based on Equation 2. Each estimate captures the differential effect of changes in global coal prices on counties with hard coal (red) and lignite (blue) endowments, relative to non-endowed counties. The dependent variables are county-level employment in total (excluding mining), manufacturing, services, and agriculture. All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.

county-level labor markets responded to coal booms, holding fixed local characteristics and national trends.[36]

Figure 5 presents the estimated effects of global coal price shocks on sectoral employment in counties with hard coal (red) and lignite (blue) deposits, relative to non-endowed counties. An increase in global coal prices leads to a significant increase in total non-mining employment in hard coal regions relative to non-coal regions, suggesting that booms boosted local labor demand. These gains are concentrated in the service sector, consistent with the hypothesis that mining activity generates local demand spillovers. Manufacturing employment also responds marginally significantly in the short run, but the effect is smaller than in services. This likely reflects adjustment frictions and the delayed responsiveness of capital-intensive tradable sectors. Agricultural employment declines slightly, consistent with a gradual structural shift away from farming, although the effect is not statistically significant. The magnitudes of the coefficient estimates are relatively modest. This is consistent with the nature of the shift–share design, which captures short-run marginal effects

[36] In the main specification, I use the exact global coal price in the year employment is observed. The results are robust to using a 5-year moving average of coal prices instead (see Figure A21d in the Appendix).

of price fluctuations, rather than the cumulative effects of large-scale structural transformation. In lignite regions, by contrast, price shocks are not associated with significant changes in employment in any sector. This supports the interpretation that lignite was weakly integrated into local production networks and generated limited upstream, downstream, or demand-side spillovers, consistent with the enclave nature of the industry. The absence of short-run negative effects likely reflects a temporary boost from mining-related demand, which may offset crowding-out pressures in other sectors.

Taken together, the results of the shift–share approach provide further evidence that coal booms in hard coal regions increased local labor demand, while failing to generate similar effects in lignite regions. The smaller coefficient estimates reflect the design's focus on short-run labor market responses to global price shocks, in contrast to the event-study, which captures the cumulative impact of resource extraction. Moreover, the results indicate that non-tradable sectors react more immediately to resource booms, while employment in tradable sectors adjusts more gradually, consistent with slower reallocation dynamics and higher adjustment costs. This alternative identification strategy reinforces the central role of production linkages in shaping the local effects of resource extraction.

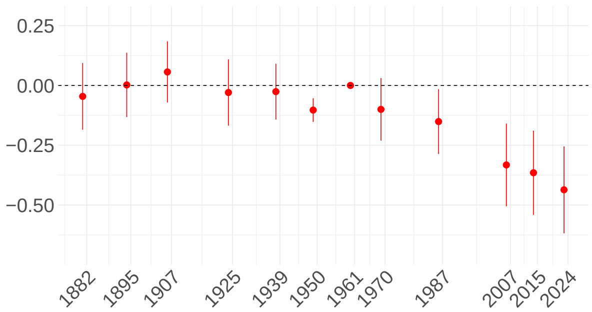
6.3 Effects of Resource Phase-Out: Event-study Results

So far, I have shown that hard coal mining generated significant economic benefits for producer regions during the extraction period. But what happens when extraction rapidly declines? Do former hard coal mining regions retain an industrial advantage through agglomeration forces, or do these benefits erode once extraction ceases?

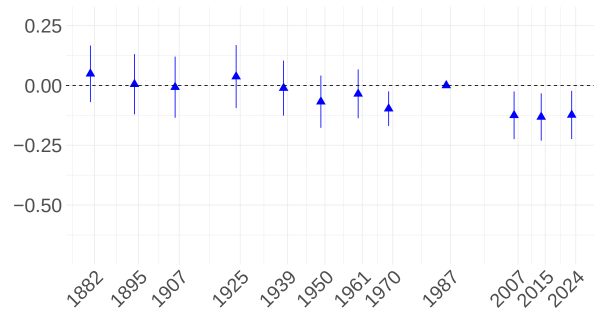
To examine the local effects of rapid mining decline, I re-estimate the event-study specification from Equation 1. I estimate separate models for hard coal and lignite to account for differences in treatment timing.[37] To isolate the respective treatment effects, I exclude the other coal type from the comparison group.[38] For the analysis of the hard coal decline, I set 1961 as the baseline

[37] Figure A9 in the Appendix shows that the main patterns are similar when both coal types are included in a single regression using 1961 as the baseline year. In this specification, the coefficients for lignite regions should be interpreted as employment dynamics over time relative to the 1961 baseline, rather than as responses to a contemporaneous phase-out.

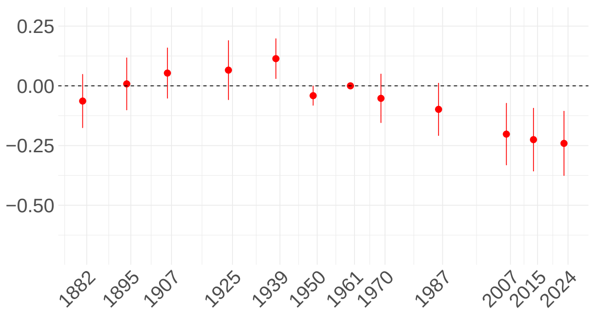
[38] Figure A7 in the Appendix shows that results are similar when the other coal region is included in the control group.



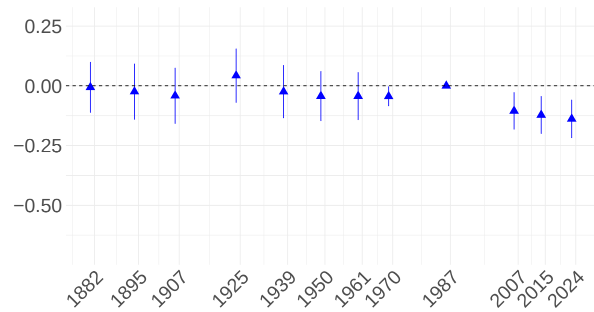
(a) Hard Coal (Manufacturing)



(b) Lignite (Manufacturing)



(c) Hard Coal (Service)



(d) Lignite (Service)

Figure 6: Event-Study Estimates: Effect of Coal Phase-Out on Manufacturing and Service Employment (1882–2024)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. The dependent variables are county-level employment in manufacturing (Panels (a) and (b)) and services (Panels (c) and (d)). All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. Each specification interacts either the hard coal (Panels (a) and (c)) or lignite (Panels (b) and (d)) treatment indicator with year fixed effects. In each case, the other coal region is excluded from the sample. The omitted base year is 1961 for hard coal and 1987 for lignite. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The sample is restricted to West German counties.

year. Around this time, hard coal counties experienced a sustained negative labor demand shock, driven by increased competition from cheap oil imports and the resulting contraction of domestic production. By contrast, the lignite decline began only in the early 1990s and remains ongoing as of 2025 (see Section 2.1). I therefore use 1987 as the baseline year. For the period before 2007, I use manufacturing employment data for all available years from 1882 onward: 1882, 1895, 1907, 1925, 1939, 1950, 1961, 1970, and 1987. From 2007 onward, annual data are available. To maintain comparability with earlier periods and avoid overrepresenting recent years, I select three reference points: 2007 (first year), 2015 (midpoint), and 2024 (latest year), thereby preserving roughly consistent spacing over time.[39]

The results indicate that the economic advantages associated with hard coal production did not persist. Figure 6 plots coefficients for manufacturing and service employment, comparing hard coal to non-coal regions in West Germany over time. The observed pre-trends across all panels support the identifying assumption of parallel trends, with no evidence of differential employment trajectories prior to 1961. In manufacturing (Panel 6a), I find a clear and persistent decline in manufacturing employment in hard coal regions relative to non-coal regions after 1961. The negative trend suggests that industrial activity in these regions eroded alongside the contraction of coal mining. By contrast, employment losses in lignite regions after 1990 are relatively small (Panel 6b). While lignite extraction is not yet fully phased out, production has declined substantially over the past three decades (see Section 2.1).[40] The comparatively modest decline in manufacturing employment reinforces the interpretation that lignite mining was only weakly integrated into local production networks, resulting in more muted labor market impacts as activity declined.

Panels 6c and 6d show similar patterns in the service sector. Hard coal regions exhibit a relative decline in service employment after 1961 compared to non-coal regions. This decline is likely driven by falling population and income levels, which weaken local service demand. In lignite regions, I also observe modest reductions in service employment after 1987, consistent with the contraction of mining activity and associated demand spillovers.

To assess the magnitude of the long-run effects of the hard coal phase-out, Table A8 in the Appendix reports results from a difference-in-differences regression. The specification replaces the event-time indicators with a single post-treatment indicator. I define the post-treatment period as beginning in 1961 for hard coal regions and in 1987 for lignite regions. The estimates indicate that manufacturing employment in hard coal regions declined by 0.22 (log) sd relative to non-coal regions after 1961. In levels, this corresponds to an average decline of approximately 21%. This suggests that the phase-out had substantial and persistent negative effects on local industrial employment. In contrast, manufacturing employment in lignite regions did not decline significantly after 1990

[39]Figure A11 in the Appendix shows that the results are robust to including all available post-2007 years (2007–2024), confirming that the choice of reference points does not drive the findings.

[40]Because both hard coal and lignite regions received targeted support during their respective phase-outs, the observed employment losses likely understate the true local impact of coal decline (Deutscher Bundestag, 2020; Röhl, 2019). Absent such interventions, the employment effects may have been more negative.

relative to non-coal regions.[41]

Figure A8 in the Appendix presents results for total employment (excluding mining) and agricultural employment. Total non-mining employment in hard coal regions exhibits a persistent decline after the onset of the coal phase-out (Panel A8a). Lignite regions also display a downward trend after 1987, although the magnitude is considerably smaller (Panel A8b). In the agricultural sector (Panels A8c and A8d), I find little evidence of significant employment changes in either type of coal region relative to non-coal areas following the substantial decline in coal extraction.

To analyze the net effect of both coal extraction and its subsequent decline, I estimate a long-run event-study covering the full period from 1849 to 2024, shown in Figure A10 in the Appendix. The analysis is restricted to counties in the intersection of modern-day Germany and historical Prussia.[42] The results show the event-study coefficients for hard coal regions in manufacturing and services converge to zero relative to the 1849 baseline by 2024. The coefficient for total employment remains marginally positive. This pattern suggests that the industrial gains accumulated during the coal extraction period were subsequently largely eroded during the phase-out. In contrast, lignite regions exhibit persistently negative coefficients through 2024 in manufacturing, services, and total employment. These negative effects appear long-lasting and are not attributable to the more recent decline in lignite production after 1990.

Taken together, my findings show that the economic benefits of hard coal production eroded as extraction declined. Rather than leaving behind a lasting industrial advantage, former hard coal regions experienced a relative contraction in manufacturing and service employment following the phase-out. Lignite regions were less affected by the decline in extraction. Employment levels in lignite regions remain persistently below those in non-coal regions relative to the pre–large-scale mining baseline.

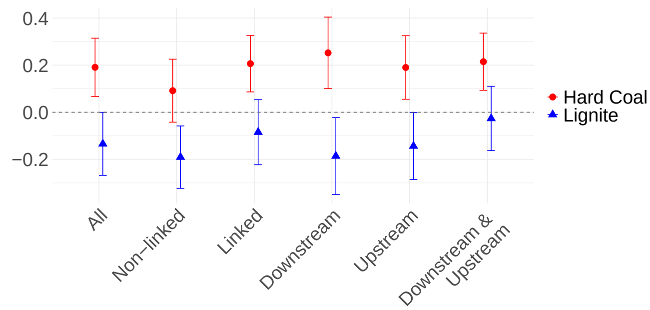
6.4 Understanding the Divergent Effects of Hard Coal and Lignite Mining

Heterogeneity by Production Linkages. To better understand the mechanisms behind the divergent economic effects of hard coal and lignite documented in the previous section, I disaggre-

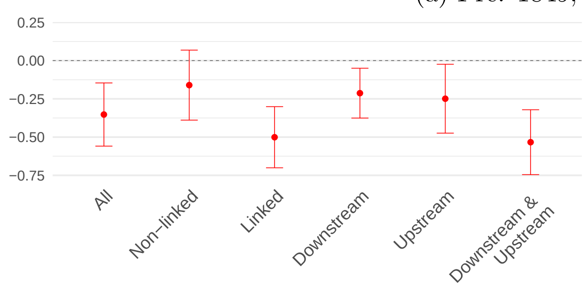
[41] As discussed in Section 2.1, the lignite phase-out did not begin until the early 1990s and remains ongoing as of 2025. Accordingly, the lignite coefficient should be interpreted as capturing the effects of a sharp contraction in mining activity, rather than the effects of a completed phase-out.

[42] I exclude census years between 1950 and 1987, as employment data for this period are only available for West Germany.

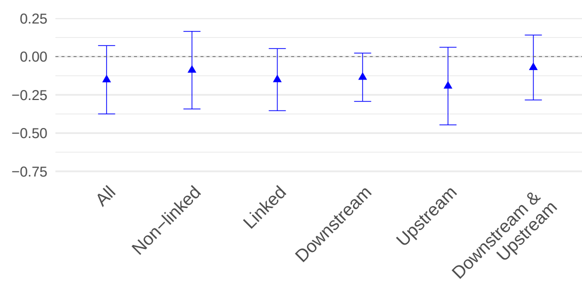
gate the manufacturing sector by the degree of upstream or downstream linkage to coal mining (see Section 4.1). For clarity and comparability, I estimate a difference-in-differences (DiD) specification based on Equation 1, in which the event-time indicators are replaced by a single post-period indicator. This enables a direct comparison of average employment changes across industries with varying degrees of production linkages to mining.



(a) Pre: 1849; Post: 1882–1939



(b) Hard Coal; Pre: 1882–1939; Post: 2007–2024



(c) Lignite; Pre: 1882–1939; Post: 2007–2024

Figure 7: Difference-in-Differences Estimates: Effect of Coal Endowment on Disaggregated Manufacturing Employment

Note: This figure plots point estimates and 95% confidence intervals from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The outcome variable is manufacturing employment, disaggregated by linkage to coal mining, classified using the U.S. 1947 Input-Output Table (U.S. Bureau of Economic Analysis, 2019). Outcomes are log-transformed and standardized to have a mean of zero and a sd of one. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Panel (a) shows results for the coal extraction period using counties within Prussian borders. The post-period includes all years from 1882 to 1939, with 1849 as the baseline year. Panels (b) and (c) show results for the coal phase-out period using counties in West Germany. The post-period includes all years from 2007 to 2024, with 1882 to 1939 as the pre-period. Panel (b) reports DiD estimates of the effect of hard coal endowment on manufacturing employment, excluding lignite regions from the sample. Panel (c) reports corresponding estimates for lignite endowment, excluding hard coal regions. Standard errors are clustered at the county level.

Panel 7a of Figure 7 shows estimates for the extraction period (1849–1939). In hard coal regions, manufacturing employment rose significantly after the onset of large-scale coal mining.

Employment gains were more pronounced in industries with stronger input–output linkages to coal. Employment effects in non-linked industries are small and statistically insignificant. By contrast, lignite regions experienced overall declines in manufacturing employment, with smaller losses in linked industries. This pattern suggests that lignite mining did not generate substantial manufacturing linkages and was instead associated with local industrial decline. Disaggregating further, the positive employment effects in hard coal regions are observed across all types of linked industries, those connected to mining through upstream linkages, downstream linkages, or both. This suggests that hard coal mining supported both local input suppliers and coal-using industries. In lignite regions, by contrast, employment effects remain weak or negative across all linkage types. For industries linked both upstream and downstream, cost pressures and spillovers appear to offset each other, resulting in an estimate close to zero.

Turning to the shift–share estimates reported in Figure A22a in the Appendix, where manufacturing employment is disaggregated by industry linkages to coal, a broadly similar pattern emerges: In hard coal regions, global price shocks are associated with positive employment responses in manufacturing sectors linked to coal. Effects in non-linked industries are negative (marginally significant). In lignite regions, by contrast, there is no meaningful employment response to coal price shocks across either sector type.

Panels 7b and 7c of Figure 7 present estimates for the coal phase-out period (1882–2024).[43] Here, the earlier patterns reverse. In hard coal regions (Panel 7b), the sharpest employment declines occur in industries that had previously been linked to mining. I find negative and significant effects for both upstream- and downstream-only linked industries. Downstream industries likely suffered because converting coal-fired steam or coking systems to oil or gas required major capital investments in new infrastructure, such as refineries, pipelines, and new heating or combustion technologies. Oil- and gas-based technologies were more readily adopted in other parts of the country, particularly in coastal regions that were not locked into coal infrastructure and had easier access to imported oil. Upstream supplier industries were likewise affected. Many specialized in producing machinery, mining equipment, and transport systems tailored to the hard coal industry. As domestic hard coal extraction declined, demand for these inputs collapsed, and technological

[43] The specification defines the pre-period using benchmark years from 1882 to 1939, and the post-period as 2007–2024. Due to data limitations, disaggregated manufacturing employment by industry is unavailable between 1950 and 1987.

reorientation toward oil or gas entailed substantial redesign and capital investment. Industries with both upstream and downstream linkages declined the most. Tightly integrated into the coal-based economy, they were doubly exposed, hit by disruptions to both input supply and output demand. Non-linked industries did not experience significant declines. In lignite regions (Panel 7c), by contrast, there are no significant differences across industry groups. This pattern aligns with lignite's weaker integration into local industrial networks.

Sectoral Concentration of Manufacturing Employment. A potential alternative explanation for the sharp decline in hard coal regions after the phase-out is that these areas had developed narrow, undiversified manufacturing structures, leaving them exposed to sector-specific shocks. To assess this, I examine changes in local industry concentration using the Herfindahl–Hirschman Index (HHI) of manufacturing employment (see Section 4.1). As shown in Figure A18 in the Appendix, the opposite pattern emerges: during the extraction period, hard coal regions became more diversified across manufacturing industries than non-coal areas. This likely reflects the broad input–output linkages of hard coal mining, which supported a wide range of supplier and user industries. By contrast, lignite regions saw a slight increase in industrial concentration over time, likely driven by crowding-out effects that constrained the development of a more diverse industrial base. These results reinforce the interpretation that the vulnerability of hard coal regions was not driven by a narrow industrial structure, but by dependence on a single leading sector. It was the depth of integration with coal, rather sectoral concentration of manufacturing employment, that amplified long-run exposure to structural decline.

Taken together, my findings show that hard coal mining spurred local industrial growth through strong production linkages, while lignite mining generated few spillovers and contributed to deindustrialization. Yet the same linkages that supported growth in hard coal regions also deepened their exposure to decline, amplifying employment losses as mining contracted.

7 Robustness of the Results

To demonstrate that my results are not sensitive to the specific choice of variables, samples, or models, I perform several robustness checks.

Balanced Subsamples. A potential concern is that changes in county boundaries over time could bias results if sample composition is systematically related to coal endowments. To address this, I re-estimate the event-study specifications (Equation 1) using a balanced panel of counties consistently observed across the full time period. Figure A5 in the Appendix presents results for the extraction period (1849–1939). While standard errors are somewhat larger, the direction and magnitude of the effects remain consistent with the baseline estimates from Figure 4. Figure A6 in the Appendix shows results for the coal phase-out period (1882–2024) using a balanced sample, with manufacturing employment as the outcome. The estimates closely mirror those from the baseline specification in Figure 6. Overall, these findings confirm that the main results are not driven by sample composition or administrative boundary changes.

Assessing Potential Pre-trends. The event-study model in Equation 1 relies on the assumption that, absent coal extraction, regions with and without deposits would have followed similar economic trends, conditional on fixed effects and observed controls. To assess the validity of this assumption, I test for differential pre-trends using two alternative outcomes: wages and urban population.

First, I re-estimate Equation 1 using unskilled wages as the outcome variable. The wage series is based on A.-K. Becker and Hornung (2025), who digitize unskilled wage data for 667 Prussian localities from 1800 to 1879 and extend the series to the county level through 1914 (see Section A.4 in the Appendix for details). I control for local price levels to account for cost-of-living differences (A.-K. Becker & Hornung, 2025). Figure A14 in the Appendix plots the estimated effects of hard coal and lignite endowments on real wages from 1800 to 1914. The results show no evidence of differential wage trends between coal and non-coal regions before 1849, supporting the parallel trends assumption. Starting in the mid-19th century, wages began to rise in hard coal regions, consistent with increasing labor demand driven by expanding mining and industrial activity. These patterns are consistent with demand-side effects from resource extraction and reinforce the interpretation that the observed employment trends are not driven by differential pre-treatment dynamics.

Second, I re-estimate Equation 1 using log urban population as the outcome variable (see Section A.4 in the Appendix for details). Figure A15 in the Appendix plots the estimated effects of hard coal and lignite endowments on city population from 1816 to 1939. The results show no evidence of differential urban population trends between coal and non-coal regions prior to 1849, supporting the parallel trends assumption. Following the onset of industrial-scale coal extraction, urban population in hard coal regions began to diverge upward, consistent with employment spillovers from mining and industrialization. In lignite regions, by contrast, no significant or sustained increases in urban population are observed. These patterns again reinforce the interpretation that the post-1849 effects are driven by resource extraction rather than pre-existing growth trends.

Together, these results provide evidence against differential growth endowments between hard coal, lignite, and non-coal regions, supporting the validity of the identifying assumption. Moreover, the use of urban population as an alternative proxy for regional development reinforces the finding that hard coal regions benefited from large-scale coal extraction, while lignite regions did not.

Potential Confounding Factors. To address concerns that the estimated effects of coal mining reflect pre-existing regional differences, I conduct a series of robustness checks that control for key geographic, institutional, and historical factors. Table A13 in the Appendix reports results from a series of difference-in-differences regressions based on Equation 1 for the coal extraction period (1849–1939), using manufacturing employment as the outcome. The specification replaces event-time indicators with a single post-period indicator. Additional controls are added sequentially to assess the robustness of the estimated effects. Column (2) adds distance to Wittenberg, a commonly used proxy for historical Protestant influence, linked to differences in human capital and institutional development (S. O. Becker & Woessmann, 2009). Column (3) includes distance to the Polish border to account for possible variation in labor supply and cross-border migration. Column (4) controls for log distance to Berlin, reflecting potential advantages from proximity to the capital such as access to political influence or public investment. Column (5) adds a west-of-Elbe indicator to capture Germany's historical east–west divide in agrarian structure and labor institutions (Tilly & Kopsidis, 2020). Finally, Column (6) includes a dummy for counties located in territories formerly occupied by France, accounting for long-run institutional effects of Napoleonic reforms (Acemoglu et al., 2011). Across all specifications, the estimated effects of coal exposure on manufacturing employment remain remarkably stable in both magnitude and statistical significance. This reinforces the interpretation that the main results are not confounded by pre-existing regional differences in geography, institutions, or human capital.

Assessing Potential Institutional Differences. To account for time-varying institutional differences across regions that could bias the estimates, I re-estimate the difference-in-differences specification for the coal extraction period (1849–1939), using manufacturing employment as the outcome. The specification replaces event-time indicators with a single post-period indicator. I include fixed effects for districts (*Regierungsbezirke*) interacted with the post-period treatment indicator. This absorbs district-specific shocks or reforms that may have differentially affected hard coal, lignite, and non-coal regions after the onset of large-scale coal extraction. The results reported in Column (5) of Table A14 in the Appendix remain robust to this specification.

Potential Contamination Bias. A potential concern is that estimating the effects of hard coal and lignite exposure in a single regression may blur their distinct effects. To address this, I re-estimate the event-study model from Equation 1 for the coal extraction period (1849–1939), separately for counties with hard coal and lignite endowments. Figures A16 and A17 in the Appendix plot the coefficient estimates. The results closely mirror the baseline estimates reported in Figure 4, confirming that the observed effects are not driven by contamination across treatment types. As a further robustness check, I exclude all counties containing both hard coal and lignite deposits. Figure A12 in the Appendix presents event-study estimates based on this restricted sample. The results remain consistent in both magnitude and significance, suggesting that the main findings are not driven by overlapping exposure.

Sensitivity to Outlier Regions and Timing of Extraction. Another concern is that the results may be driven by a disproportionately large coal-producing district. To address this, I re-estimate the difference-in-differences specifications based on Equation 1 for the coal extraction period (1849–1939), using manufacturing employment as the outcome. The specification replaces event-time indicators with a single post-period indicator. I exclude key outliers from the sample. Specifically, I drop the largest hard coal-producing district (Arnsberg) and the largest lignite-producing district (Merseburg).[44] As shown in Columns (2) and (4) of Table A14 in the Appendix, the results remain robust in both magnitude and significance, indicating that the main findings are not driven by high-output districts.

[44]Based on total production between 1850 and 1914. Own calculations using data from Kunz (2008).

A related concern is that the positive economic effects observed in hard coal regions may reflect their earlier onset of extraction rather than structural differences in linkage intensity (see Figure A1a in the Appendix). To address this, I re-estimate the difference-in-differences specifications excluding Oppeln, the earliest major hard coal-producing district.[45] As shown in Column (3) of Table A14 in the Appendix, the results remain robust in both magnitude and statistical significance. For lignite, Merseburg was both the earliest and largest producer. Column (4) reports the corresponding estimates excluding this district, which closely mirror the baseline results.

Together with the finding that employment gains in hard coal regions were concentrated in sectors with strong upstream and downstream linkages, these results support the interpretation that the observed divergence between hard coal and lignite regions reflects differences in economic integration, rather than differences in extraction timing or the influence of outlier regions.

Alternative Definitions of Coal Exposure. A concern is that the estimated effects may be sensitive to how exposure to coal deposits is defined. In the baseline specification, I classify a county as exposed to hard coal or lignite if a deposit lies within a 20 kilometer radius of the county centroid. To test the robustness of the geographic treatment definition, I re-estimate the difference-in-differences specification based on Equation 1 for the coal extraction period (1849–1939), using manufacturing employment as the outcome and replacing event-time indicators with a single post-period indicator. I vary the distance threshold for coal endowment from the baseline 20 km to alternative cutoffs of 10, 15, 25, and 30 km. As shown in Table A12 in the Appendix, the results remain qualitatively consistent across specifications. The baseline 20-kilometer radius neither maximizes the estimated effect of hard coal nor minimizes the negative effect of lignite. The estimated effect of hard coal exposure is statistically significant at distances up to 20 km, but attenuates and becomes insignificant beyond that range. This spatial pattern suggests that the economic benefits of hard coal extraction were concentrated near production sites, consistent with the clustering of coal-linked industries and strong, localized input–output linkages. In contrast, the negative effects of lignite exposure remain statistically significant even at wider radii, suggesting that the adverse economic impacts of extraction extended into neighboring regions.

Additionally, I restrict the analysis to coal deposits located within Prussia. The results, reported

[45]Based on production levels in 1850. Own calculations using data from Kunz (2008).

in Column (2) of Table A11 in the Appendix, remain robust. Finally, I replace the binary treatment indicators with continuous measures of proximity to the nearest hard coal or lignite deposit, allowing treatment intensity to vary with distance. As shown in Column (3) of Table A11 in the Appendix, the results align with the baseline: manufacturing employment increases with proximity to hard coal and decreases with proximity to lignite. Overall, these findings show that the main results are not driven by the specific geographic definition of coal exposure and hold across a range of alternative treatment measures.

Potential Spatial Correlation in Standard Errors. To account for potential spatial autocorrelation in the error structure, I re-estimate the baseline event-study for the coal extraction period using Conley (1999) standard errors, which allow for correlation of residuals across geographic space. In addition, I follow S. O. Becker et al. (2025) and correct for spatial unit roots. The results, shown in Columns (1) and (2) of Figure A13 in the Appendix using manufacturing employment as the outcome variable, remain marginally significant across most years.

Spatial Spillovers. Next, I test for geographic spillovers. Following Butts (2021), I extend the difference-in-differences model to include indicators for counties located 20–25 km and 25–30 km from the nearest coal deposit, These concentric distance bands capture indirect effects in adjacent areas and help assess the robustness of the main estimates.

Table A9 in the Appendix presents results for the extraction period (1849–1939). The increase in manufacturing employment in hard coal-producing counties remains robust to the inclusion of spillover terms, though the magnitude declines slightly. Neighboring counties, particularly those in the 25–30 km ring, experience employment declines, consistent with factor reallocation toward mining hubs. For lignite, employment falls not only in producing counties but also in adjacent areas, suggesting broader negative spillovers.

Columns (1)–(3) of Table A10 in the Appendix report estimates for the hard coal phase-out period, using 1961 as the treatment date. Spillover terms are introduced sequentially. The core treatment effect remains negative and statistically significant across all specifications. Coefficients for neighboring zones are small and statistically insignificant. This suggests that employment losses during the phase-out were spatially concentrated in producer counties. Columns (4)–(6) of

Table A10 in the Appendix report estimates for the lignite decline, using 1987 as the treatment date. The results indicate negative but statistically insignificant effects in both lignite-producing and adjacent counties, with marginal significance in the 20–25 km ring. These effects are relatively modest compared to the more pronounced impacts observed for the hard coal phase-out.

Taken together, hard coal extraction generated localized employment gains, while lignite extraction led to diffuse regional losses. The subsequent hard coal phase-out produced spatially concentrated employment declines, whereas the lignite decline was associated with weaker and more dispersed negative effects.

Alternative Time-Series Variation To assess the robustness of the shift–share estimates, I test whether the results are sensitive to alternative definitions of the time-varying shift component. In the baseline specification, the shift is given by the world coal price, which captures global demand and supply shocks in the coal market and is plausibly exogenous to local economic conditions in Germany. This helps address potential endogeneity concerns that may arise when using national variables, which could partially reflect local outcomes in coal-producing regions. As a robustness check, however, I re-estimate the shift–share model using coal-type-specific, rather than aggregate, indicators of coal sector activity. Specifically, I use several alternative national-level variables: German hard coal and lignite production, prices, and revenue (price times quantity).[46] These alternatives allow for a clearer differentiation between hard coal and lignite dynamics. Moreover, I use a five-year moving average of the world coal price to account for potential lags in the adjustment of local economic conditions to global market shocks.

As shown in Figure A21 in the Appendix, the results remain consistent across specifications in both magnitude and significance. This confirms that the estimated effects are not sensitive to the precise choice of the shift variable.

To assess whether the estimated effects are driven by unrelated global trends rather than coal-specific price dynamics, I conduct placebo regressions in which global coal prices are replaced with

[46]Using coal revenues as the shift variable is particularly informative, as it captures both supply-side movements (quantities) and demand-side movements (prices) that determine the mining sector's overall scale. Quantity-only measures may understate differences between hard coal and lignite, since lignite has a lower unit value, even when extraction costs per unit of energy are similar. Conversely, price-only measures can be misleading, as prices may fluctuate due to supply disruptions, or trade restrictions, even when production remains stable. By combining both prices and quantities, production value offers a more comprehensive and economically meaningful proxy for the intensity of coal sector activity at the national level.

prices of unrelated commodities (coffee, sugar, tobacco, and rice). The estimated interaction coefficients for hard-coal and lignite regions are near zero or slightly negative (Figure A22b in the Appendix), providing no evidence of analogous effects and reinforcing the validity of the identification strategy.

Alternative Input-Output Table and Linkage Definition. To ensure that the observed heterogeneity by linkage intensity is not driven by the specific input–output table used, I replicate the analysis using an alternative classification. Specifically, I use the historical input–output table from Fremdling and Staeglin (2014), which is based on 1936 data for Germany. This table enables me to disaggregate the industries in my occupation panel into 25 manufacturing sectors. Industries are reclassified by their upstream and downstream linkage intensity using this table (see Table A5 in the Appendix). Then, I re-estimate the difference-in-differences models for both the coal extraction and the coal phase-out period. As shown in Figure A19 in the Appendix, the results remain highly consistent in both magnitude and significance, confirming that the findings are robust to the choice of input–output classification.

In addition, I assess whether the observed effects depend on the linkage definition. Figure A20 in the Appendix replicates the analysis for the coal extraction period using direct linkages only (left panel) and second-degree (indirect) linkages (right panel) for both upstream and downstream relationships, based on the U.S. 1947 input–output table (U.S. Bureau of Economic Analysis, 2019). The results remain robust to these alternative specifications.

Together, these robustness checks confirm the validity of my results across both the extraction and the phase-out period, and reinforce the heterogeneity patterns.

8 Conclusion

This paper offers an explanation for why similar industries can lead regions down sharply divergent development paths, depending on the strength of local production linkages. My results suggest that industries are more likely to generate lasting local benefits when they create strong connections to the broader economy. Where an industry remains isolated, regions tend to stagnate or decline.

I compare two coal industries in Germany that operated under similar institutional conditions but differed in their integration into local supply chains. The results show that a resource sector's ability to generate upstream and downstream linkages is a key mechanism shaping its broader local economic impact.

My findings show that regions specialized in hard coal mining, an industry with strong input–output linkages, experienced significant employment gains during the extraction period, with spillovers into both manufacturing and services. Manufacturing gains were more pronounced in industries with upstream or downstream linkages to coal mining. Hard coal regions also became more diversified across manufacturing sub-sectors over time. This indicates that resource-based growth did not lead to industrial concentration. However, after the coal phase-out, hard coal regions suffered persistent employment losses, especially in sectors tightly connected to mining. My results show that this vulnerability was not due to a lack of diversification across industries. Instead, it was the depth of integration with the coal economy that amplified the costs of decline. Results from a long-run event-study spanning 1849 to 2024 suggest that the initial employment gains in hard coal regions, driven by large-scale extraction, were largely offset by subsequent losses during the phase-out. In contrast, lignite regions, where extraction is more isolated from the local economy, saw adverse effects even during extraction. These included deindustrialization and a decline in local services. Yet their weaker integration into industrial supply chains made them less exposed to economic disruptions during the subsequent decline in lignite production. Taken together, my findings highlight the double-edged nature of production linkages: while they may amplify growth when a major industry expands, they may also amplify losses when it contracts. This mechanism helps explain why regions exposed to seemingly similar industries can experience sharply divergent development paths.

My findings contribute to ongoing debates on place-based and industrial policy in advanced economies. Regional disparities have been widening in recent decades, raising concerns about political stability and social cohesion (D. Autor et al., 2019; Pierce & Schott, 2020; Rodríguez-Pose, 2018). To address this, governments devote substantial resources to place-based policies aimed at supporting distressed regions. In the United States, such programs cost nearly 50 billion USD annually (T. J. Bartik, 2020). In the European Union, almost a third of the total EU budget for 2021–2027 has been allocated to cohesion policy to reduce regional disparities and support lagging areas (European Commission, 2021). Assumptions about how industrial expansion and contraction affect local labor markets play a central role in the design and targeting of spatial policies.

I show that targeting tradable sectors with high local linkage potential can generate larger local multipliers than supporting enclaved industries with limited integration into the regional economy. In some cases, support for such isolated sectors may even crowd out other activities and hinder broader development. My results suggest that designing place-based industrial policy around local production linkages can promote regional development and help narrow spatial disparities. At the same time, when a major industry begins to decline, the very linkages that once supported growth can transmit negative shocks across the local economy. Without effective transition strategies, regions with tightly embedded industrial structures may face deeper employment losses and more severe deindustrialization. This risk underscores the importance of forward-looking adjustment policies. These are especially important during structural transformations such as decarbonization or digitization, where the phase-out of core industries is foreseeable. Lastly, this paper helps reconcile conflicting findings in the resource literature. It sheds light on why the legacy of resource wealth can range from long-run prosperity to persistent underdevelopment.

References

Acemoglu, D., Cantoni, D., Johnson, S., & Robinson, J. A. (2011). The consequences of radical reform: The french revolution. *American Economic Review*, *101*(7), 3286–3307.

Acemoglu, D., Carvalho, V. M., Özdaglar, A., & Tahbaz-Salehi, A. (2012). The network origins of aggregate fluctuations. *Econometrica*, *80*(5), 1977–2016.

Adao, R., Kolesár, M., & Morales, E. (2019). Shift-share designs: Theory and inference. *The Quarterly Journal of Economics*, *134*(4), 1949–2010.

Allcott, H., & Keniston, D. (2018). Dutch disease or agglomeration? The local economic effects of natural resource booms in modern America. *The Review of Economic Studies*, *85*(2), 695–731.

Allgemeines Berggesetz für die Preußischen Staaten, Berlin (1865, June 24). Retrieved October 19, 2025, from https://www.lwl.org/westfaelische-geschichte/que/normal/que1088.pdf

Aragón, F. M., & Rud, J. P. (2013). Natural resources and local communities: Evidence from a Peruvian gold mine. *American Economic Journal: Economic Policy*, *5*(2), 1–25.

Asch, K., & Bellenberg, S. (2005). *The 1: 5 million international geological map of Europe and adjacent areas (IGME 5000)*. Bundesanstalt für Geowissenschaften und Rohstoffe.

Ashraf, Q. H., Cinnirella, F., Galor, O., Gershman, B., & Hornung, E. (2025). Structural change, elite capitalism, and the emergence of labour emancipation. *Review of Economic Studies*, *92*(2), 808–836.

Austin, B. A., Glaeser, E. L., & Summers, L. H. (2018). *Jobs for the heartland: Place-based policies in 21st century America* (NBER Working Paper No. 24548). National Bureau of Economic Research.

Autor, D., Dorn, D., & Hanson, G. (2019). When work disappears: Manufacturing decline and the falling marriage market value of young men. *American Economic Review: Insights*, *1*(2), 161–178.

Autor, D., Dorn, D., Hanson, G., & Majlesi, K. (2020). Importing political polarization? The electoral consequences of rising trade exposure. *American Economic Review*, *110*(10), 3139–3183.

Autor, D. H., Dorn, D., & Hanson, G. H. (2016). The china shock: Learning from labor-market adjustment to large changes in trade. *Annual Review of Economics*, *8*(1), 205–240.

Banken, R. (1993). Die Diffusion der Dampfmaschine in Preußen um 1830. *Jahrbuch für Wirtschaftsgeschichte*, *34*(2), 219–248.

Barrot, J.-N., & Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *The Quarterly Journal of Economics*, *131*(3), 1543–1592.

Bartels, C., Kersting, F., & Wolf, N. (2025). Testing marx: Capital accumulation, income inequality, and socialism in late nineteenth-century Germany. *Review of Economics and Statistics*, 1–16.

Bartik, A. W., Currie, J., Greenstone, M., & Knittel, C. R. (2019). The local economic and welfare consequences of hydraulic fracturing. *American Economic Journal: Applied Economics*, *11*(4), 105–155.

Bartik, T. J. (1991). Who benefits from state and local economic development policies? *W.E. Upjohn Institute for Employment Research.*

Bartik, T. J. (2020). Using place-based jobs policies to help distressed communities. *Journal of Economic Perspectives*, *34*(3), 99–127.

Becker, A.-K., & Hornung, E. (2025). Industrialization and the return to labor: Evidence from Prussia. *CEPR Discussion Paper*, (20810).

Becker, S. O., Boll, P. D., & Voth, H.-J. (2025). *Spatial unit roots in regressions: A practitioner's guide and a stata package* (CESifo Working Paper Series No. 11651). CESifo. https://EconPapers.repec.org/RePEc:ces:ceswps:_11651

Becker, S. O., & Woessmann, L. (2009). Was Weber Wrong? A Human Capital Theory of Protestant Economic History. *The Quarterly Journal of Economics*, *124*(2), 531–596.

Berbée, P., Braun, S. T., & Franke, R. (2025). Reversing fortunes of german regions, 1926–2019: Boon and bane of early industrialization? *Journal of Economic Growth*, *30*(2), 307–337.

Berman, N., Couttenier, M., Rohner, D., & Thoenig, M. (2017). This mine is mine! how minerals fuel conflicts in Africa. *American Economic Review*, *107*(6), 1564–1610.

Black, D., McKinnish, T., & Sanders, S. (2005). The economic impact of the coal boom and bust. *The Economic Journal*, *115*(503), 449–476.

Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of Economic Studies*, *89*(1), 181–213.

Bundesagentur für Arbeit. (2025). *Sozialversicherungspflichtig Beschäftigte (SvB) am Arbeitsort: Deutschland, Kreise (Gebietsstand Mai 2025)* (Statistische Auswertung No. 388767). Düsseldorf.

Bundesberggesetz, Bonn (1980, August 13). Retrieved October 19, 2025, from https://www.gesetze-im-internet.de/bbergg/

Butts, K. (2021). Difference-in-differences with spatial spillovers. *arXiv preprint arXiv:2105.03737*.

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230.

Carvalho, V. M. (2014). From micro to macro via production networks. *Journal of Economic Perspectives*, *28*(4), 23–48.

Caselli, F., & Michaels, G. (2013). Do oil windfalls improve living standards? Evidence from brazil. *American Economic Journal: Applied Economics*, *5*(1), 208–238.

Cavalcanti, T., Da Mata, D., & Toscani, F. (2019). Winning the oil lottery: The impact of natural resource extraction on growth. *Journal of Economic Growth*, *24*(1), 79–115.

Châtel & Dollfus. (1931). *Atlas Châtel et Dollfus: Les houillères européennes*. Société de Documentation Industrielle.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, *92*(1), 1–45.

Corden, W. M., & Neary, J. P. (1982). Booming sector and de-industrialisation in a small open economy. *The economic journal*, *92*(368), 825–848.

Criscuolo, C., Martin, R., Overman, H. G., & Van Reenen, J. (2019). Some causal effects of an industrial policy. *American Economic Review*, *109*(1), 48–85.

Cust, J., & Poelhekke, S. (2015). The local economic impacts of natural resource extraction. *Annual Review of Resource Economics*, *7*(1), 251–268.

Dal Bó, E., & Dal Bó, P. (2011). Workers, warriors, and criminals: Social conflict in general equilibrium. *Journal of the European Economic Association*, *9*(4), 646–677.

De Pleijt, A., Nuvolari, A., & Weisdorf, J. (2020). Human Capital Formation During the First Industrial Revolution: Evidence from the use of Steam Engines. *Journal of the European Economic Association*, *18*(2), 829–889.

Dechezleprêtre, A., & Sato, M. (2017). The impacts of environmental regulations on competitiveness. *Review of Environmental Economics and Policy*, *11*(2), 183–206.

Deutscher Bundestag. (2020). *Strukturmaßnahmen zum Kohleausstieg der Länder Brandenburg, Sachsen, Sachsen-Anhalt und Thüringen* (Dokumentation WD 5-3000-006/20). Wissenschaftliche Dienste des Deutschen Bundestage. Berlin. Retrieved November 17, 2025, from https://www.bundestag.de/resource/blob/683762/WD-5-006-20-pdf.pdf

Deutscher Bundestag. (2024). *Die Bergrechtliche Grundabtretung in der Rechtsentwicklung seit dem Allgemeinen Berggesetz für die Preußischen Staaten* (Sachstand WD 5–3000–002/24). Wissenschaftliche Dienste des Deutschen Bundestages. Retrieved October 19, 2025, from https://www.bundestag.de/resource/blob/992662/WD-5-002-24-pdf.pdf

Dippel, C., Gold, R., Heblich, S., & Pinto, R. (2022). The effect of trade on workers and voters. *The Economic Journal*, *132*(641), 199–217.

Dix-Carneiro, R., & Kovak, B. K. (2017). Trade liberalization and regional dynamics. *American Economic Review*, *107*(10), 2908–2946.

Dube, O., & Vargas, J. F. (2013). Commodity price shocks and civil conflict: Evidence from Colombia. *The Review of Economic Studies*, *80*(4), 1384–1421.

Eggert, U. (1883). Die Bewegung der Holzpreise und Tagelohn-Sätze in den preussischen Staatsforsten von 1800 bis 1879. *Zeitschrift des Königlich Preussischen Statistischen Bureaus*, *23*, 1–44.

Ellison, G., & Glaeser, E. L. (1997). Geographic concentration in US manufacturing industries: A dartboard approach. *Journal of Political Economy*, *105*(5), 889–927.

Ellison, G., Glaeser, E. L., & Kerr, W. R. (2010). What Causes Industry Agglomeration? Evidence from Coagglomeration Patterns [citation key:]. *American Economic Review*, *100*(3), 1195–1213.

Esposito, E., & Abramson, S. F. (2021). The European coal curse. *Journal of Economic Growth*, *26*(1), 77–112.

European Commission. (2021). Cohesion policy 2021–2027. Retrieved November 13, 2025, from https://ec.europa.eu/regional_policy/en/2021_2027/

Fernihough, A., & O'Rourke, K. H. (2021). Coal and the European industrial revolution. *The Economic Journal*, *131*(635), 1135–1149.

Feyrer, J., Mansur, E. T., & Sacerdote, B. (2017). Geographic dispersion of economic shocks: Evidence from the fracking revolution. *American Economic Review*, *107*(4), 1313–1334.

Fick, S. E., & Hijmans, R. J. (2017). WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, *37*(12), 4302–4315.

Fischer, W. (2011). *Statistik der Bergbauproduktion Deutschlands 1850–1914*. GESIS Data Archive.

Franck, R., & Galor, O. (2021). Flowers of evil? Industrialization and long run development. *Journal of Monetary Economics*, *117*, 108–128.

Fremdling, R. (1995). Anglo-German rivalry on coal markets in France, the Netherlands and Germany, 1850-1913.

Fremdling, R., & Staeglin, R. (2014). An input-output table of germany in 1936: A documentation of results, sources and research strategy. *Jahrbuch für Wirtschaftsgeschichte*, *55*(2), 187–298.

Fritzsche, M., & Wolf, N. (2023). *Fickle fossils. economic growth, coal and the european oil invasion, 1900-2015* (No. 10805). CESifo Working Paper.

Gagliardi, L., Moretti, E., & Serafinelli, M. (2023). *The world's rust belts: The heterogeneous effects of deindustrialization on 1,993 cities in six countries* (Working Paper No. 31948). National Bureau of Economic Research.

Galor, O., & Özak, Ö. (2016). The agricultural origins of time preference. *American Economic Review*, *106*(10), 3064–3103.

Glaeser, E. L. (2008). *Cities, agglomeration, and spatial equilibrium*. OUP Oxford.

Glaeser, E. L., Kerr, S. P., & Kerr, W. R. (2015). Entrepreneurship and urban growth: An empirical assessment with historical mines. *Review of Economics and Statistics*, *97*(2), 498–520.

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, *110*(8), 2586–2624.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277.

Greenstone, M., Hornbeck, R., & Moretti, E. (2010). Identifying agglomeration spillovers: Evidence from winners and losers of large plant openings. *Journal of Political Economy*, *118*(3), 536–598.

Gunton, T. (2003). Natural resources and regional development: An assessment of dependency and comparative advantage paradigms. *Economic Geography*, *79*(1), 67–94.

Gutberlet, T. (2014). Mechanization and the spatial distribution of industries in the German Empire, 1875 to 1907. *The Economic History Review*, *67*(2), 463–491.

Havranek, T., Horvath, R., & Zeynalov, A. (2016). Natural resources and economic growth: A meta-analysis. *World Development*, *88*, 134–151.

Hirschman, A. O. (1958). *The strategy of economic development*. Yale University Press.

Hooker, M. A., & Knetter, M. M. (2001). Measuring the economic effects of military base closures. *Economic Inquiry*, *39*(4), 583–598.

Hornung, E. (2015). Railroads and Growth in Prussia. *Journal of the European Economic Association*, *13*(4), 699–736.

Hovis, L., & Mouat, J. (1996). Miners, engineers, and the transformation of work in the western mining industry. *Technology and Culture*, *37*(3), 429–456.

Jacks, D. S. (2019). From boom to bust: A typology of real commodity prices in the long run. *Cliometrica*, *13*(2), 201–220.

Jacobsen, G. D., & Parker, D. P. (2016). The economic aftermath of resource booms: Evidence from boomtowns in the American West. *The Economic Journal*, *126*(593), 1092–1128.

Juhász, R., Lane, N., & Rodrik, D. (2024). The new economics of industrial policy. *Annual Review of Economics*, *16*(2024), 213–242.

Kaiserliches Statistisches Amt. (1884). *Berufs- und Betriebszählung vom 5. Juni 1882: Berufsstatistik* (Vol. 4). Puttkammer & Mühlbrecht.

Kaiserliches Statistisches Amt. (1897). *Berufs- und Betriebszählung vom 14. Juni 1895: Berufsstatistik* (Vol. 109). Puttkammer & Mühlbrecht.

Kaiserliches Statistisches Amt. (1910). *Berufs- und Betriebszählung vom 12. Juni 1907: Berufsstatistik* (Vol. 209). Puttkammer & Mühlbrecht.

Kline, P., & Moretti, E. (2014). People, places, and public policy: Some simple welfare economics of local economic development programs. *Annual Review of Economics*, *6*(1), 629–662.

Kohleverstromungsbeendigungsgesetz (KVBG). Gesetz zur Reduzierung und zur Beendigung der Kohleverstromung und zur Änderung weiterer Gesetze (Kohleausstiegsgesetz) (2020). Retrieved October 22, 2025, from https://www.gesetze-im-internet.de/kvbg/

Krugman, P. (1987). The narrow moving band, the Dutch disease, and the competitive consequences of Mrs. Thatcher. *Journal of Development Economics*, *27*(1-2), 41–55.

Kunz, A. (Ed.). (2008). *HGIS Germany: Historical information system of the german states (1820–1914)*. Mainz, Institut für Europäische Geschichte (IEG). Retrieved October 8, 2025, from http://www.ekomp.digihist.de/Dokumentation_Datensaetze/Zeitreihen/Bergbau

Lane, N. (2025). Manufacturing revolutions: Industrial policy and industrialization in south korea. *The Quarterly Journal of Economics*, qjaf025.

Lippert, A. (2014). *Spill-overs of a resource boom: Evidence from Zambian copper mines*. University of Oxford.

Marshall, A. (1890). *Principles of Economics* (8th). Macmillan.

Matzerath, H. (1985). *Urbanisierung in Preußen 1815–1914*. Kohlhammer / Deutscher Gemeindeverlag.
Michaels, G. (2011). The long term consequences of resource-based specialisation. *The Economic Journal*, *121*(551), 31–57.
Moretti, E., & Thulin, P. (2013). Local multipliers and human capital in the United States and Sweden. *Industrial and Corporate Change*, *22*(1), 339–362.
Moretti, E. (2010). Local Multipliers. *American Economic Review*, *100*(2), 373–377.
Moretti, E. (2011). Local labor markets. In O. Ashenfelter & D. E. Card (Eds.), *Handbook of Labor Economics*. Elsevier.
Morris, M., Kaplinsky, R., & Kaplan, D. (2012). "One thing leads to another"– Commodities, linkages and industrial development. *Resources Policy*, *37*(4), 408–416.
Neumark, D., & Simpson, H. (2015). Place-based policies. In *Handbook of regional and urban economics* (pp. 1197–1287). Elsevier.
Papyrakis, E., & Raveh, O. (2014). An empirical analysis of a regional dutch disease: The case of Canada. *Environmental and Resource Economics*, *58*(2), 179–198.
Pelzl, P., & Poelhekke, S. (2021). Good mine, bad mine: Natural resource heterogeneity and dutch disease in Indonesia. *Journal of International Economics*, *131*, 103457.
Peters, M. (2022). Market size and spatial growth—evidence from Germany's post-war population expulsions. *Econometrica*, *90*(5), 2357–2396.
Pierce, J. R., & Schott, P. K. (2020). Trade liberalization and mortality: Evidence from US counties. *American Economic Review: Insights*, *2*(1), 47–63.
Pierenkemper, T. (1992). Die schwerindustriellen Regionen Deutschlands in der Expansion: Oberschlesien, die Saar und das Ruhrgebiet im 19. Jahrhundert. *Jahrbuch für Wirtschaftsgeschichte/Economic History Yearbook*, *33*(1), 37–56.
Preußisches Gewerbesteuergesetz vom 24. Juni 1891 (1891). Retrieved October 23, 2025, from https://www.jstor.org/stable/40904491?seq=1
Rodríguez-Pose, A. (2018). The revenge of the places that don't matter (and what to do about it). *Cambridge Journal of Regions, Economy and Society*, *11*(1), 189–209.
Röhl, K.-H. (2019). Das Ruhrgebiet: der anhaltende industrielle Strukturwandel im Spiegel der Regionalpolitik. *Wirtschaftsdienst*, *99*(Sonderheft 13), 49–55.
Rosenthal, S. S., & Strange, W. C. (2001). The determinants of agglomeration. *Journal of Urban Economics*, *50*(2), 191–229.
Sachs, J. D., & Warner, A. M. (2001). The curse of natural resources. *European Economic Review*, *45*(4-6), 827–838.
Schmitz, J. (1888). *Übersicht der für die sämtlichen deutschen Bundesstaaten in Gemäßheit des § 8 des Reichsgesetzes betreffend die Krankenversicherung der Arbeiter vom 15. Juni 1883 festgestellten ortsüblichen Tagelöhne gewerblicher Tagearbeiter* (2nd ed.). Heuser's Verlag.
Shao, S., Zhang, Y., Tian, Z., Li, D., & Yang, L. (2020). The regional dutch disease effect within China: A spatial econometric investigation. *Energy Economics*, *88*, 104766.
Sheremirov, V., & Spirovska, S. (2022). Fiscal multipliers in advanced and developing countries: Evidence from military spending. *Journal of Public Economics*, *208*, 104631.
Siegloch, S., Wehrhöfer, N., & Etzel, T. (2025). Spillover, efficiency, and equity effects of regional firm subsidies. *American Economic Journal: Economic Policy*, *17*(1), 144–180.
Statista Research Department. (2025a). *Braunkohle: Förderung in Deutschland seit 1990*. Retrieved September 29, 2025, from https://de.statista.com/statistik/daten/studie/156258/umfrage/braunkohlefoerderung-in-deutschland-seit-1990/

Statista Research Department. (2025b). *Steinkohle: Förderung in Deutschland seit 1970.* Retrieved September 29, 2025, from https://de.statista.com/statistik/daten/studie/161462/umfrage/steinkohle-foerderung-in-deutschland-seit-1970/

Statistisches Bundesamt. (1953). *Berufs-, Volks- und Betriebszählung vom 13. September 1950: Berufszählung* (Vol. 101).

Statistisches Bundesamt. (1964). *Volks- und Berufszählung vom 6. Juni 1961: Berufszählung* (Vol. 215).

Statistisches Bundesamt. (1972). *Volks- und Berufszählung vom 27. Mai 1970: Berufszählung* (Vol. 301).

Statistisches Bundesamt. (1989). *Volks-, Berufs- und Betriebszählung vom 25. Mai 1987: Berufszählung* (Vol. 401).

Statistisches Bureau zu Berlin. (1851–1855). *Tabellen und amtliche nachrichten über den preussischen staat für das jahr 1849* (Vol. 1–6b). A. W. Hayn.

Statistisches Reichsamt. (1929). *Volks-, Berufs- und Betriebszählung vom 16. Juni 1925: Berufszählung* (Vol. 403–405). Verlag für Sozialpolitik, Wirtschaft und Statistik.

Statistisches Reichsamt. (1934). Vierteljahresstatistiken über Erzeugung, Verbrauch, Einfuhr und Versand von Kohle. Bundesarchiv (BArch), Bestand R 3102: Reichswirtschaftsministerium, Signatur R 3102/3545, Band 1, Berlin-Lichterfelde.

Statistisches Reichsamt. (1936a). Produktionserhebung 1936 zum gesamten Bergbau und zur Erdölgewinnung. Bundesarchiv (BArch), Bestand R 3102: Reichswirtschaftsministerium, Signatur R 3102/3709, Band 2, Berlin-Lichterfelde.

Statistisches Reichsamt. (1936b). *Volks-, Berufs- und Betriebszählung vom 16. Juni 1933: Berufszählung* (Vol. 453). Verlag für Sozialpolitik, Wirtschaft und Statistik.

Statistisches Reichsamt. (1943). *Volks-, Berufs- und Betriebszählung vom 17. Mai 1939: Berufszählung* (Vol. 557–559).

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, *225*(2), 175–199.

Tilly, R. H., & Kopsidis, M. (2020). *From old regime to industrial state: A history of German industrialization from the eighteenth century to World War I.* University of Chicago Press.

U.S. Bureau of Economic Analysis. (2019). Summary-level use table before redefinitions, 1947–1962. Retrieved October 29, 2024, from https://apps.bea.gov/

van der Ploeg, F. (2011). Natural resources: Curse or blessing? *Journal of Economic Literature*, *49*(2), 366–420.

Walker, W. R. (2013). The transitional costs of sectoral reallocation: Evidence from the clean air act and the workforce. *The Quarterly Journal of Economics*, *128*(4), 1787–1835.

Watkins, M. H. (1963). A staple theory of economic growth. *Canadian Journal of Economics and Political Science*, *29*(2), 141–158.

Zentralblatt. (1892). *Zentralblatt für das Deutsche Reich* (Reichsamt des Inneren, Ed.; Vol. 20-42). Carl Lehmanns Verlag.

A Appendix

A.1 Additional Tables

A.1.1 Descriptives

Table A1: Descriptive Statistic: Cross-Sectional Data

Variable	N	Mean	SD	Min	Max
Panel (a): Treatment Variables					
Distance Hard coal (in km)	1122	132	111	0	552
Distance Lignite (in km)	1122	110	110	0	566
Hard coal region (dummy)	1122	0.16	0.36	0	1
Lignite region (dummy)	1122	0.18	0.39	0	1
Hard coal or lignite region (dummy)	1122	0.3	0.46	0	1
Hard coal and lignite region (dummy)	1122	0.03	0.18	0	1
Panel (b): Geographic Control Variables					
Distance River (in km)	1122	13	18	0	113
Distance Coast (in km)	1122	223	140	0	455
Caloric Suitability	1122	1667	274	714	2313
Slope	1122	1.2	1.1	0.03	12
Temperature	1122	8.5	0.84	5.8	11
Rainfall	1122	63	15	42	143

Note: This table reports summary statistics for key cross-sectional variables used in the main analyses. Units of observation are counties as of 1900. N refers to the number of counties.

Table A2: Descriptive Statistic: Pooled Panel Data

Variable	N	Mean	SD	Min	Max
Panel A: 1849–1939					
Total Employment	2365	31592	54052	3775	1774040
Manufacturing Employment	2365	10327	28129	415	908951
Service Employment	2365	7419	26494	163	849575
Agricultural Employment	2365	12686	8605	84	61594
Panel B: 1882–1939					
Total Employment	5100	23995	42357	677	1774040
Manufacturing Employment	5100	9056	21882	240	908951
Service Employment	5100	6193	20442	97	849575
Agricultural Employment	5100	8169	5549	15	38107
Panel C: 1882–2024					
Total Employment	7642	29583	45025	768	1774040
Manufacturing Employment	7642	11896	21208	385	908951
Service Employment	7642	12367	26731	123	849575
Agricultural Employment	7642	4804	4807	0	32285

Note: This table reports summary statistics for key panel variables used in the main analyses. Panel (a) covers all Prussian counties from 1849 to 1939. Panel (b) covers counties within the borders of the German Empire from 1882 to 1939 (balanced panel). Panel (c) covers counties in West Germany from 1882 to 2024. Total employment excludes mining. N refers to the number of observations.

Table A3: Descriptive Statistic: Time-Series Data (1882–1939)

Variable	N	Mean	SD	Min	Max
World Coal Price	5100	93	7.3	83	104
World Coal Price (5Y Avg.)	5100	99	11	86	116
Hard Coal Production	5100	109	48	46	186
Lignite Production	5100	65	48	12	141
Hard Coal Price	5100	9.6	3.5	5.1	15
Lignite Price	5100	2.6	0.24	2.4	3
Hard Coal Revenue	5100	1148	705	236	2130
Lignite Revenue	5100	165	119	36	335

Note: This table reports summary statistics for key time-series variables used in the main analysis, based on counties in the German Empire from 1882 to 1939. N refers to the number of observations.

Table A4: Upstream and Downstream Linkage Measures for U.S. Input–Output Sectors (1947)

	Upstream		Downstream		Linked		
Industry	Direct	Indirect	Direct	Indirect	Downstream	Upstream	Total
Apparel and leather and allied products	0.0	0.2	0.1	0.6	No	No	No
Chemical products	1.5	0.4	1.8	1.2	No	Yes	Yes
Fabricated metal products	1.0	0.7	1.7	5.8	Yes	Yes	Yes
Food and beverage and tobacco products	0.0	0.2	0.1	0.6	No	No	No
Machinery	2.9	0.7	0.3	3.7	Yes	Yes	Yes
Miscellaneous manufacturing	0.1	0.6	2.9	2.6	Yes	No	Yes
Motor vehicles, bodies and trailers, and parts	0.7	0.4	0.2	2.9	Yes	No	No
Nonmetallic mineral products	0.9	0.5	15.2	1.8	Yes	Yes	Yes
Paper products	0.3	0.4	1.9	1.0	No	No	Yes
Petroleum and coal products	1.2	0.8	5.1	2.4	Yes	Yes	Yes
Plastics and rubber products	0.2	0.9	0.7	1.2	No	No	No
Primary metals	0.8	0.7	14.4	1.1	Yes	Yes	Yes
Printing and related support activities	0.3	0.6	0.1	1.5	No	No	No
Textile mills and textile product mills	0.0	0.4	0.3	0.5	No	No	No
Wood products	2.1	0.4	0.4	1.6	No	Yes	Yes

Note: This table reports direct and indirect upstream (sales to mining) and downstream (inputs from mining) linkage shares in percent for U.S. manufacturing and service sectors, based on the 1947 summary-level Use table (before redefinitions) (U.S. Bureau of Economic Analysis, 2019). I compute linkage shares on the full I-O matrix, but report only those industries that can be matched to historical occupations, and I exclude own-use (diagonal). Consequently, the reported column sums for upstream and downstream shares (direct and indirect) need not equal one. Binary "linked" indicators are defined as specified in Section 4.1.

Table A5: Upstream and Downstream Linkage Measures for German Input–Output Sectors (1936)

	Upstream		Downstream		Linked		
Industry	Direct	Indirect	Direct	Indirect	Downstream	Upstream	Total
Building and construction	0.3	1.3	0.4	2.8	No	Yes	Yes
Chemical industry	2.4	0.4	4.6	5.2	Yes	Yes	Yes
Chemical-technical industry	0.0	1.2	0.0	4.1	No	No	Yes
Clothing	0.0	1.3	0.0	1.1	No	Yes	Yes
Edible oil and fats	0.0	0.1	0.0	2.0	No	No	No
Electrical engineering	0.4	1.0	0.5	4.0	Yes	Yes	Yes
Electricity, gas and water	4.0	0.7	33.4	3.0	Yes	Yes	Yes
Fabricated iron and steel products	0.6	0.9	0.0	6.8	Yes	Yes	Yes
Food, beverages and tobacco	0.0	0.7	0.1	2.4	No	No	Yes
Foundries	6.4	0.4	0.0	6.8	Yes	Yes	Yes
Fuel industries	0.0	1.2	18.8	2.6	Yes	No	Yes
Glass	0.0	0.6	0.0	10.2	Yes	No	No
Leather industry	0.0	0.2	0.0	2.0	No	No	No
Machinery	0.0	1.6	0.0	4.4	No	Yes	Yes
Manufacture of paper and paper products	0.0	0.4	0.0	5.2	Yes	No	No
Manufactured wood products	0.2	0.6	0.1	2.5	No	No	Yes
Metal products	0.0	1.5	0.0	5.5	Yes	Yes	Yes
Non-ferrous metals	0.9	0.5	4.8	4.1	Yes	Yes	Yes
Precision engineering, optics	0.0	1.2	0.0	3.5	No	No	No
Printing and duplicating	0.4	0.8	0.0	1.8	No	No	No
Rubber and asbestos manufacturing	0.0	1.4	0.0	5.9	Yes	Yes	Yes
Saw mills, timber processing	12.9	0.2	0.0	2.8	No	Yes	Yes
Spirits industry	0.0	0.5	0.0	1.8	No	No	No
Stone and quarrying	0.1	0.8	3.5	13.7	Yes	No	Yes
Textiles	0.1	0.1	0.0	2.8	No	No	No

Note: This table reports direct and indirect upstream (sales to mining) and downstream (inputs from mining) linkage shares in percent for German manufacturing sectors, based on the 1936 I-O table (Fremdling & Staeglin, 2014). I compute linkage shares on the full I-O matrix, but report only those industries that can be matched to historical occupations, and I exclude own-use (diagonal). Consequently, the reported column sums for upstream and downstream shares (direct and indirect) need not equal one. Binary "linked" indicators are defined as specified in Section 4.1.

A.1.2 Event-study Estimates

Table A6: Event-Study Estimates: Effect of Resource Endowment on Sectoral Employment (1849–1939)

	Total (1)	Manufacturing (2)	Service (3)	Agricultural (4)
HardCoal x 1882	0.399***	0.207***	0.126**	0.089
	(0.084)	(0.059)	(0.064)	(0.083)
HardCoal x 1895	0.466***	0.246***	0.157**	0.085
	(0.105)	(0.072)	(0.073)	(0.077)
HardCoal x 1907	0.468***	0.268***	0.188**	-0.033
	(0.119)	(0.082)	(0.081)	(0.096)
HardCoal x 1925	0.396***	0.185*	0.127	-0.087
	(0.142)	(0.098)	(0.085)	(0.096)
HardCoal x 1939	0.439***	0.166	0.154	-0.089
	(0.160)	(0.104)	(0.095)	(0.092)
Lignite x 1882	-0.086	-0.073	-0.129**	0.342***
	(0.093)	(0.058)	(0.059)	(0.091)
Lignite x 1895	-0.242**	-0.214***	-0.181**	0.312***
	(0.111)	(0.076)	(0.071)	(0.084)
Lignite x 1907	-0.343***	-0.283***	-0.245***	0.268**
	(0.127)	(0.091)	(0.080)	(0.108)
Lignite x 1925	-0.371***	-0.269***	-0.232***	0.290***
	(0.140)	(0.102)	(0.080)	(0.107)
Lignite x 1939	-0.364**	-0.192*	-0.192**	0.192*
	(0.166)	(0.111)	(0.092)	(0.113)
Geo Controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365
Adjusted R^2	0.72	0.86	0.88	0.82

Notes: This table reports coefficients from event-study regressions based on Equation 1. The dependent variables are the log of standardized county-level employment in (1) total (excluding mining), (2) manufacturing, (3) service, and (4) agricultural sectors (mean zero, unit variance). Each row presents estimated effects for counties with hard coal or lignite deposits interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors, clustered at the county level, are shown in parentheses. The analysis is restricted to Prussian counties. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

A.1.3 Difference-in-differences Estimates

Table A7: Difference-in-differences Estimates: Effect of Resource Endowment on Sectoral Employment (1849–1939)

	Total (1)	Manufacturing (2)	Service (3)	Agricultural (4)
HardCoal x Post1849	0.423***	0.208***	0.143**	-0.008
	(0.111)	(0.074)	(0.073)	(0.076)
Lignite x Post1849	-0.300***	-0.217***	-0.209***	0.275***
	(0.114)	(0.079)	(0.068)	(0.087)
Geo Controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365
Adjusted R^2	0.69	0.85	0.87	0.82

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variables are the log of standardized county-level employment in (1) total (excluding mining), (2) manufacturing, (3) services, and (4) agriculture (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal counties interacted with a post indicator equal to one for all years after 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A8: Difference-in-differences Estimates: Effect of Coal Phase-Out on Manufacturing and Service Employment (1882–2024)

	Manufacturing (1)	Service (2)	Manufacturing (3)	Service (4)
HardCoal x Post1961	-0.223***	-0.143**		
	(0.080)	(0.059)		
Lignite x Post1987			-0.089	-0.047
			(0.084)	(0.047)
Geo Controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	6,598	6,598	6,590	6,590
Adjusted R^2	0.83	0.90	0.82	0.90

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variables are the log of standardized county-level employment in manufacturing and services (mean zero, unit variance). Columns (1) and (2) report estimated effects for counties with hard coal deposits relative to non-coal counties, interacted with a post-period indicator that equals one for all years after 1961. Columns (3) and (4) report estimated effects for counties with lignite deposits relative to non-coal counties, interacted with a post-period indicator that equals one for all years after 1987. In each case, the other coal region is excluded from the sample. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to West German counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A9: Difference-in-differences Estimates: Spatial Spillovers (1849–1939)

	Baseline (1)	+ 20–25km (2)	+ 25–30km (3)
HardCoal x Post1849	0.208***	0.180**	0.155**
	(0.074)	(0.075)	(0.077)
HardCoal (20–25km) x Post1849		-0.186	-0.164
		(0.131)	(0.131)
HardCoal (25–30km) x Post1849			-0.238*
			(0.127)
Lignite × Post1849	-0.217***	-0.241***	-0.248***
	(0.079)	(0.080)	(0.081)
Lignite (20–25km) x Post1849		-0.388**	-0.415**
		(0.169)	(0.170)
Lignite (25–30km) x Post1849			-0.251**
			(0.124)
Geo Controls	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	2,365	2,365	2,365
Adjusted R^2	0.85	0.85	0.85

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal counties interacted with a post indicator equal to one for all years after 1849. Spillover terms, defined as indicators for counties located within successive 20–25 km and 25–30 km rings around hard coal or lignite deposits, are introduced sequentially. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A10: Difference-in-differences Estimates: Spatial Spillovers (1882–2024)

	Hard Coal			Lignite		
	Baseline (1)	+ 20–25km (2)	+ 25–30km (3)	Baseline (4)	+ 20–25km (5)	+ 25–30km (6)
HardCoal × Post1961	-0.223***	-0.216***	-0.215**			
	(0.080)	(0.082)	(0.083)			
HardCoal (20–25km) × Post1961		0.075	0.076			
		(0.195)	(0.196)			
HardCoal (25–30km) × Post1961			0.018			
			(0.150)			
Lignite × Post1987				-0.089	-0.101	-0.108
				(0.084)	(0.084)	(0.085)
Lignite (20–25km) × Post1987					-0.226*	-0.232*
					(0.129)	(0.129)
Lignite (25–30km) × Post1987						-0.108
						(0.149)
Geo Controls	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,598	6,598	6,598	6,590	6,590	6,590
Adjusted R^2	0.83	0.83	0.83	0.82	0.82	0.82

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Columns (1)–(3) report estimated effects for counties with hard coal deposits, and Columns (4)–(6) for counties with lignite deposits, each relative to non-coal counties. The treatment indicator equals one for all years after 1961 (Columns (1)–(3)) and after 1987 (Columns (4)–(6)). In each case, the other coal region is excluded from the sample. Spillover terms, defined as indicators for counties located within successive 20–25 km and 25–30 km rings around hard coal or lignite deposits, are introduced sequentially. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to West German counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A11: Difference-in-differences Estimates: Alternative Definitions of the Treatment Variable (1849–1939)

	Baseline	Coal Inside Prussia	Proximity
	(1)	(2)	(3)
HardCoal x Post1849	0.208***	0.185**	0.074*
	(0.074)	(0.074)	(0.042)
Lignite x Post1849	-0.217***	-0.206***	-0.158***
	(0.079)	(0.079)	(0.030)
Geo Controls	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	2,365	2,365	2,365
Adjusted R^2	0.85	0.85	0.85

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties classified as exposed to hard coal or lignite, with exposure defined in different ways, interacted with a post indicator equal to one for all years after 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A12: Difference-in-differences Estimates: Varying the Distance Cut-off (1849–1939)

	10km (1)	15km (2)	20km (3)	25km (4)	30km (5)
HardCoal x Post1849	0.242***	0.136*	0.208***	0.117	0.042
	(0.067)	(0.070)	(0.074)	(0.073)	(0.074)
Lignite x Post1849	-0.231***	-0.246***	-0.217***	-0.284***	-0.291***
	(0.079)	(0.077)	(0.079)	(0.081)	(0.076)
Geo Controls	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365	2,365
Adjusted R^2	0.85	0.85	0.85	0.85	0.85

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variables is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits, defined using varying distance cutoffs, interacted with a post indicator equal to one for all years after 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A13: Difference-in-differences Estimates: Potential Confounders (1849–1939)

	Baseline (1)	Distance Wittenberg (2)	Distance Poland (3)	Distance Berlin (4)	West Elbe (5)	French Occupation (6)
HardCoal x Post1849	0.208***	0.117*	0.121*	0.161**	0.188***	0.190***
	(0.074)	(0.068)	(0.068)	(0.068)	(0.067)	(0.067)
Lignite x Post1849	-0.217***	-0.153*	-0.160*	-0.101	-0.163**	-0.161**
	(0.079)	(0.092)	(0.092)	(0.079)	(0.081)	(0.081)
Geo Controls	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365	2,365	2,365
Adjusted R^2	0.85	0.86	0.86	0.86	0.86	0.86

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variables is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal counties interacted with a post indicator equal to one for all years after 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Potential confounders are added one at a time. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

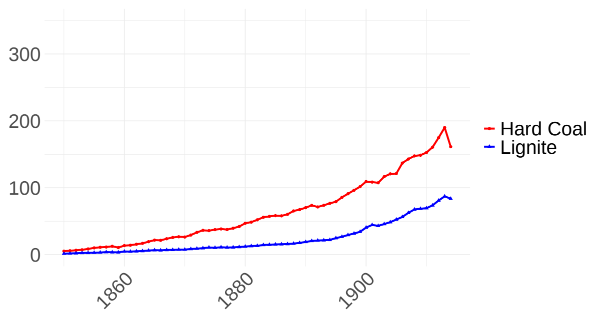
Table A14: Difference-in-differences Estimates: Specification Robustness (1849–1939)

	Baseline (1)	Largest Producer Hard coal (2)	First Mover Hard Coal (3)	First Mover = Largest Producer Lignite (4)	District x Post1849 (5)	HardCoal vs. Lignite (6)
HardCoal × Post1849	0.208***	0.131*	0.151**	0.163**	0.218***	0.257***
	(0.074)	(0.076)	(0.076)	(0.080)	(0.072)	(0.098)
Lignite × Post1849	-0.217***	-0.196**	-0.199**	-0.254***	-0.339***	
	(0.079)	(0.081)	(0.079)	(0.085)	(0.103)	
Geo Controls	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,365	2,227	2,230	2,257	2,365	1,022
Adjusted R^2	0.85	0.85	0.86	0.85	0.86	0.81

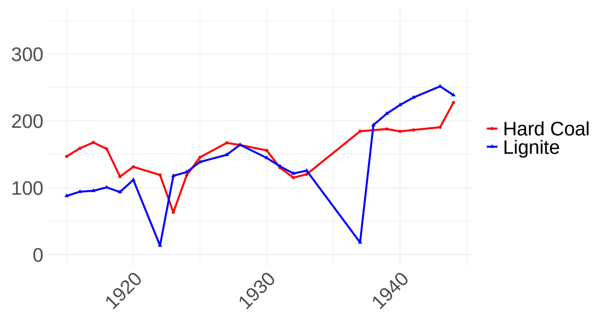
Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The dependent variables is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal counties interacted with a post indicator equal to one for all years after 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.2 Additional Figures

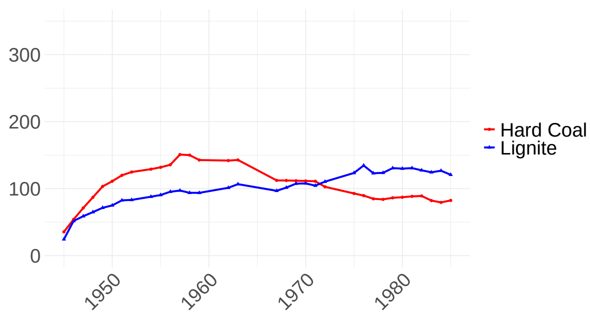
A.2.1 Descriptives



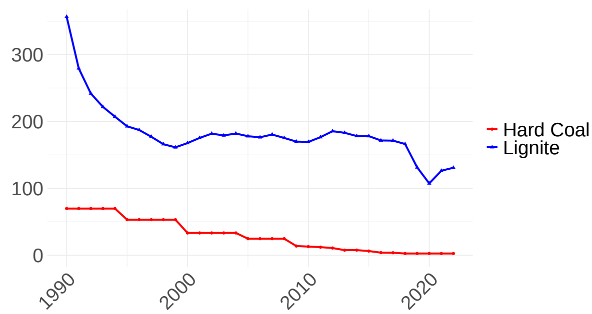
(a) 1850–1914



(b) 1915–1944



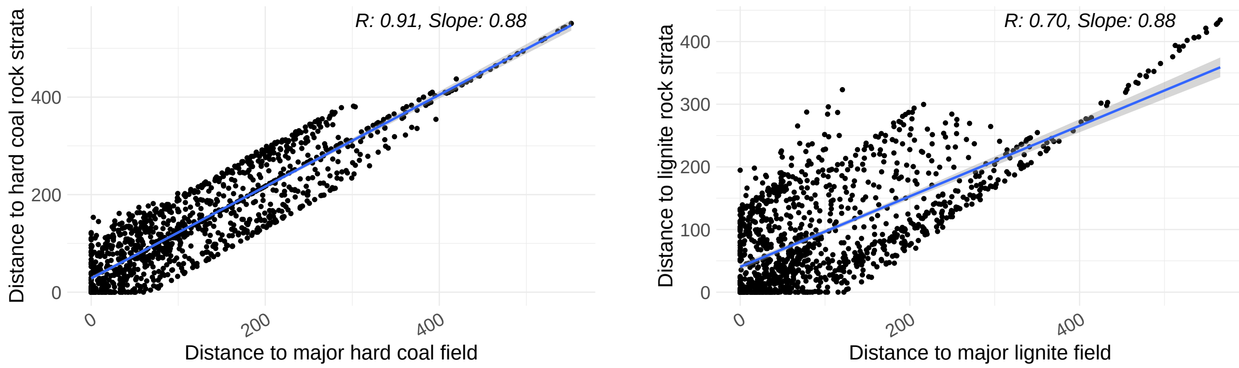
(c) 1945–1985



(d) 1990–2023

Figure A1: Hard Coal and Lignite Production over Time in Germany (1850–2023)

Note: This figure shows hard coal and lignite production in Germany over time, measured in million tons. Data for the period 1850–1985 are from Fischer (2011); data for 1990–2023 are from (Statista Research Department, 2025a, 2025b).



(a) Hard coal

(b) Lignite

Figure A2: Correlation Between Distance to Major Coal Fields and Distance to Rock Strata in Germany

Note: The scatterplots show the correlation between the centroid distance of a county to the nearest large hard coal (left) or lignite (right) field and to the nearest carboniferous or lignite rock strata, respectively (in km). Data sources: Asch and Bellenberg, 2005; Fernihough and O'Rourke, 2021.

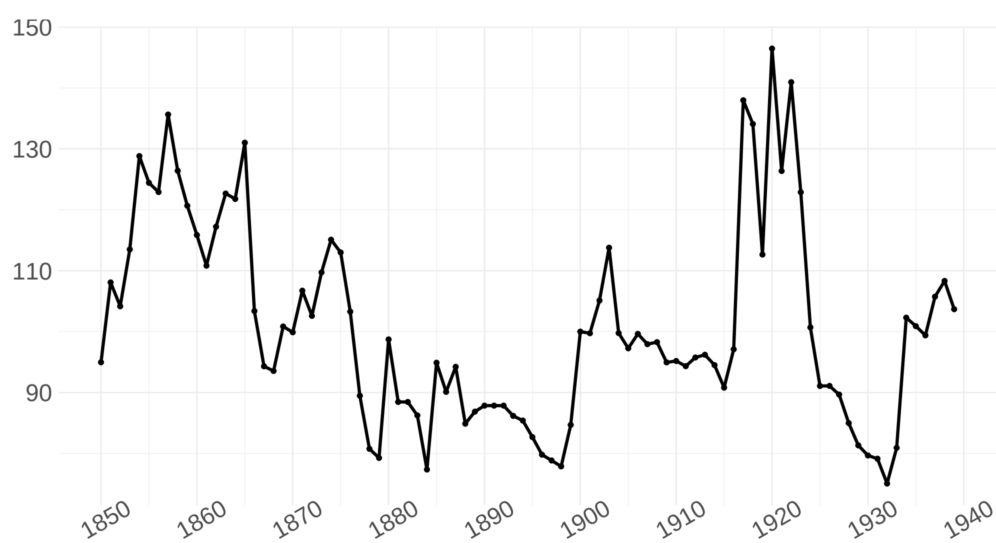
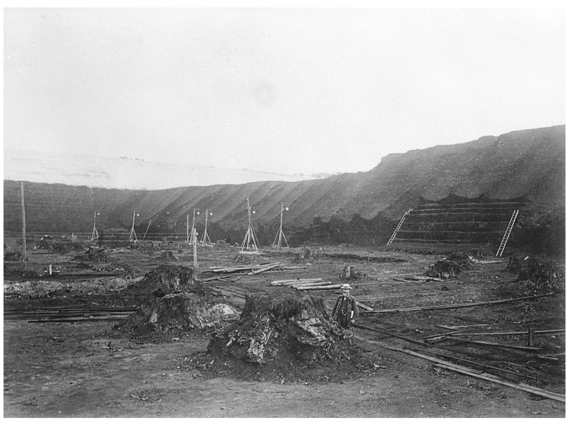


Figure A3: World Coal Prices over Time

Note: The figure plots annual world coal prices, 1850–1939, expressed in 1900 U.S. dollars. Source: Jacks (2019).



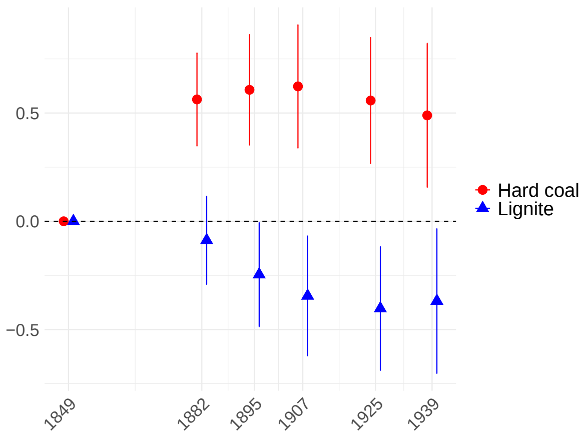
(a) Lignite Mining (1900, Senftenberg)
© Deutsche Fotothek (Franz Stoedtner)

(b) Hard Coal Mining (1893, Freital-Burgk)
© Deutsche Fotothek (Heinrich Börner)

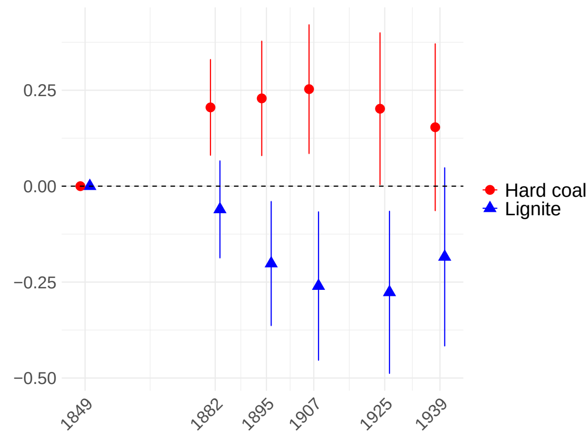
Figure A4: Illustrating Differences in Coal Extraction Methods: Hard coal and Lignite Mining

Note: Panel (a) shows a typical lignite mine in 1900, where extraction was carried out via open-pit surface mining. Panel (b) depicts hard coal mining in the late 19th century, which relied on complex underground extraction techniques, including shafts, tunnels, and ventilation systems.

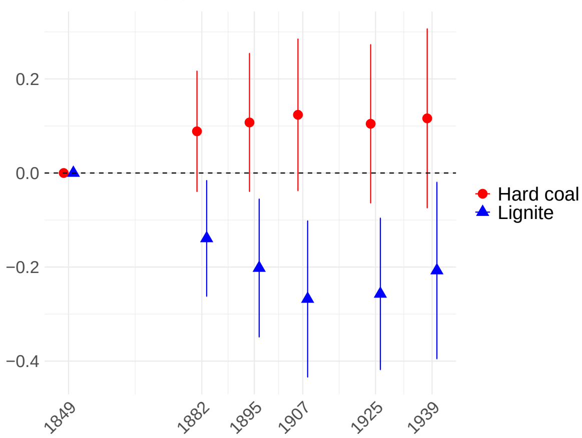
A.2.2 Event-study Estimates



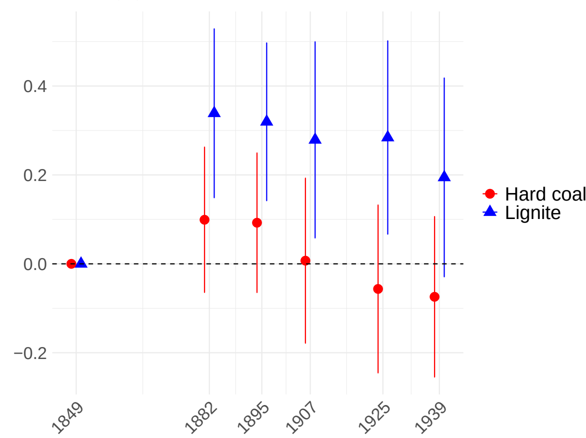
(a) Total employment



(b) Manufacturing employment



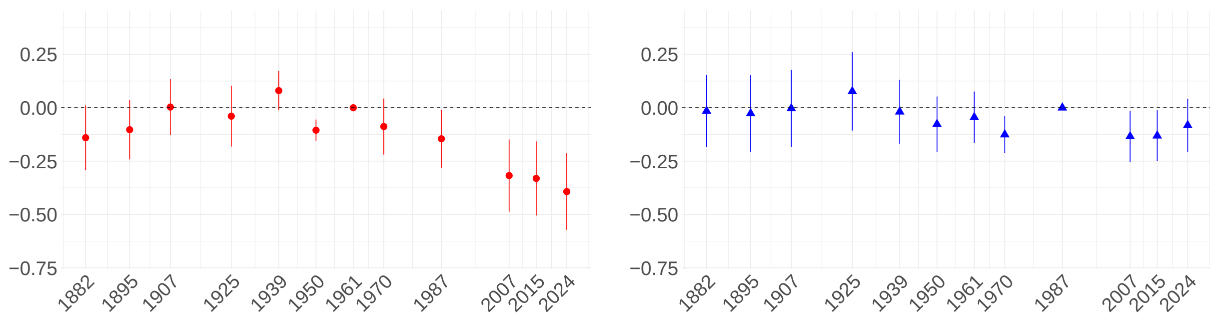
(c) Service employment



(d) Agricultural employment

Figure A5: Event-Study Estimates: Effect of Resource Endowment on Sectoral Employment (1849–1939, Balanced Sample)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1 using a balanced sample of Prussian counties. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level.

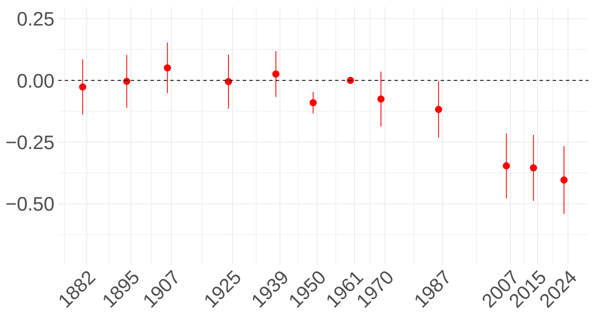


(a) Hard Coal

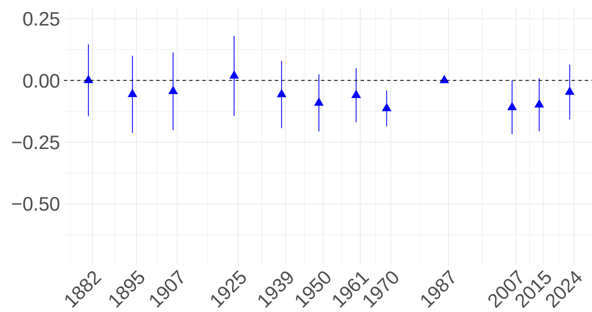
(b) Lignite

Figure A6: Event-Study Estimates: Effect of Coal Phase-Out on Manufacturing Employment (1882–2024, Balanced Sample)

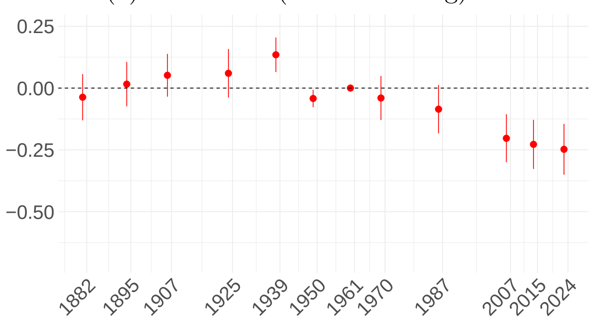
Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1 using a balanced sample. The dependent variables are county-level manufacturing employment, log-transformed and standardized to have a mean of zero and a sd of one. Each specification interacts either the hard coal (Panel (a)) or lignite (Panel (b)) treatment indicator with year fixed effects. In each case, the other coal region is excluded from the sample. The omitted base year is 1961 for hard coal and 1987 for lignite. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to West German counties.



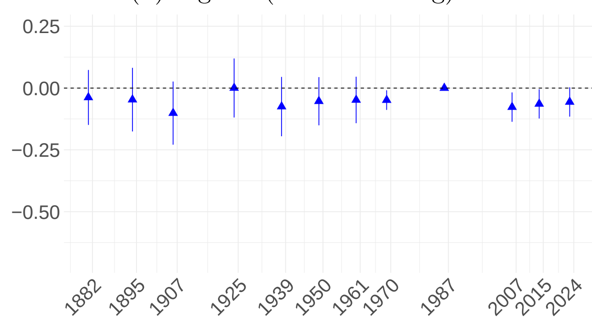
(a) Hard Coal (Manufacturing)



(b) Lignite (Manufacturing)



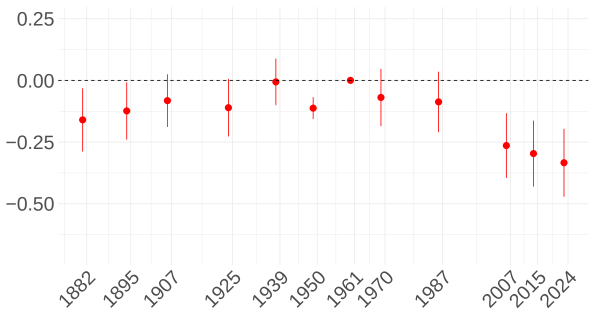
(c) Hard Coal (Service)



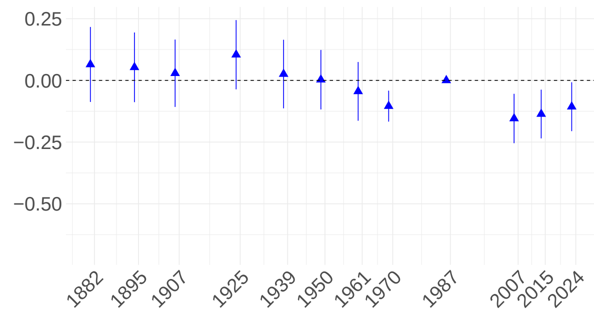
(d) Lignite (Service)

Figure A7: Event-Study Estimates: Effect of Coal Phase-Out on Manufacturing and Service Employment (1882–2024, Alternative Counterfactual)

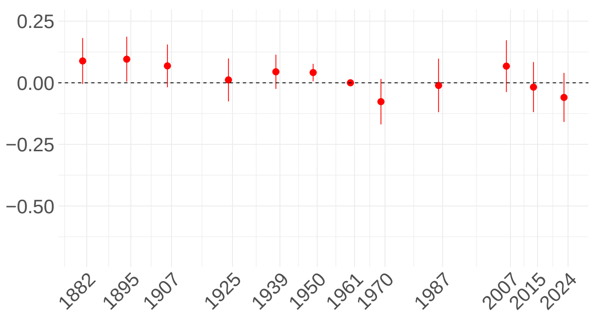
Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. The dependent variables are county-level employment in manufacturing (Panels (a) and (b)) and services (Panels (c) and (d)). All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. Each specification interacts either the hard coal (Panels (a) and (c)) or lignite (Panels (b) and (d)) treatment indicator with year fixed effects. The counterfactual group consists of non-coal regions and counties from the other coal region. The omitted base year is 1961 for hard coal and 1987 for lignite. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The sample is restricted to West German counties.



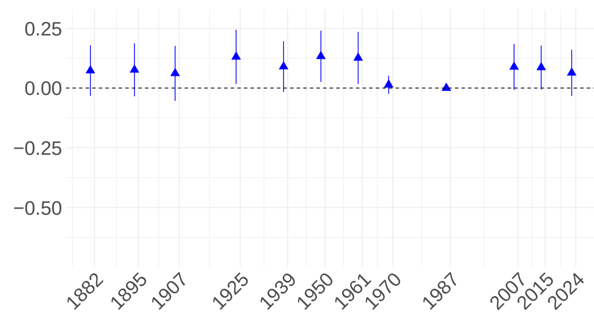
(a) Hard Coal (Total employment)



(b) Lignite (Total employment)



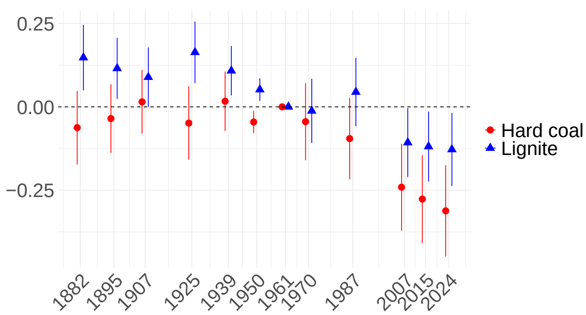
(c) Hard Coal (Agricultural)



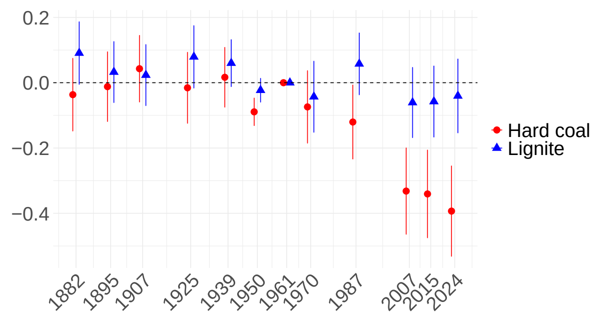
(d) Lignite (Agricultural)

Figure A8: Event-Study Estimates: Effect of Coal Phase-Out on Total and Agricultural Employment (1882–2024)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. The dependent variables are county-level total employment excluding mining (Panels (a) and (b)) and agricultural employment (Panels (c) and (d)). All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. Each specification interacts either the hard coal (Panels (a) and (c)) or lignite (Panels (b) and (d)) treatment indicator with year fixed effects. In each case, the other coal region is excluded from the sample. The omitted base year is 1961 for hard coal and 1987 for lignite. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The sample is restricted to West German counties.

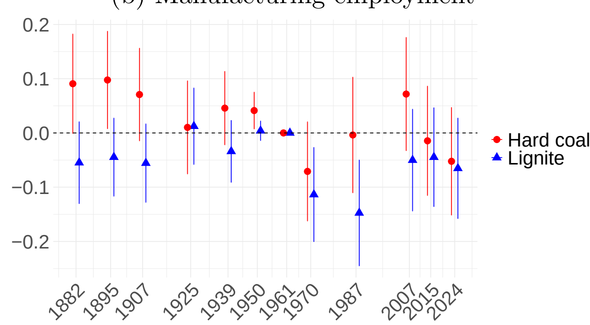


(a) Total employment



(b) Manufacturing employment

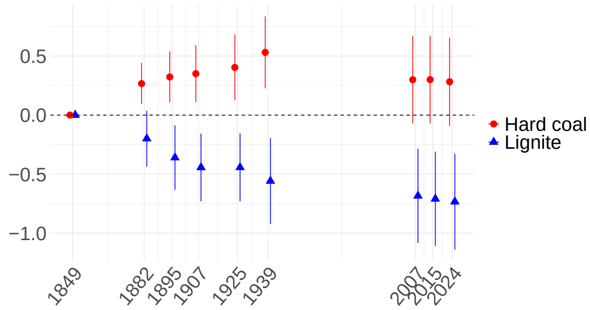
(c) Service employment



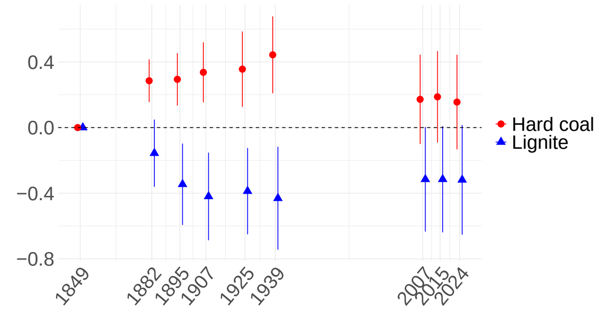
(d) Agricultural employment

Figure A9: Event-Study Estimates: Effect of Coal Phase-Out on Sectoral Employment (1882–2024), Including Hard Coal and Lignite

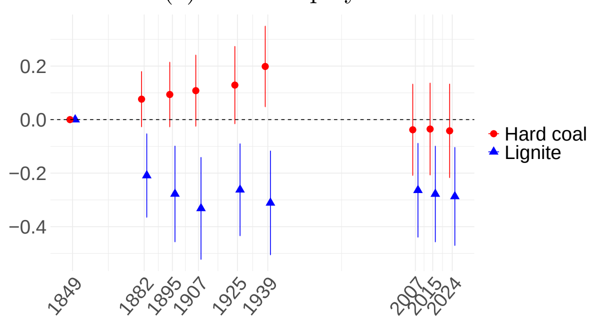
Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite indicators are interacted with year fixed effects. The omitted year is 1961. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to West German counties.



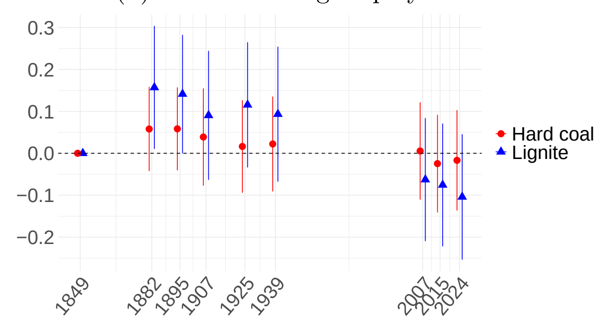
(a) Total employment



(b) Manufacturing employment



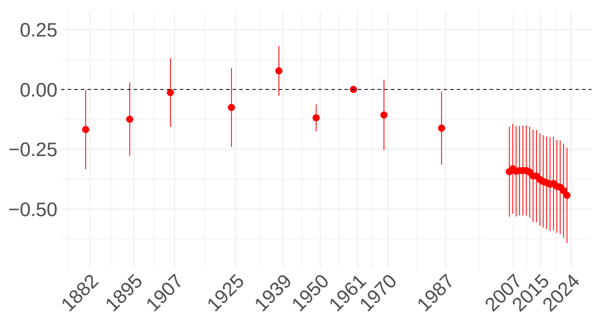
(c) Service employment



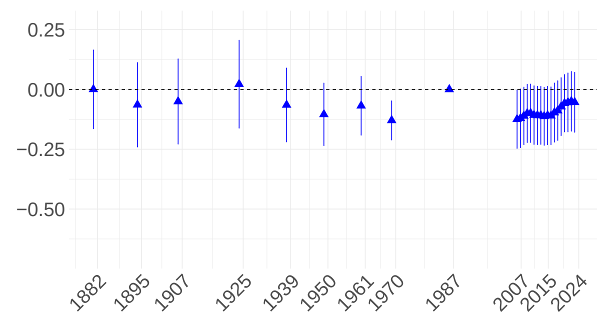
(d) Agricultural employment

Figure A10: Event-Study Estimates: Effect of Resource Endowment on Sectoral Employment (1849–2024)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to counties in the intersection of modern-day Germany and historical Prussia.



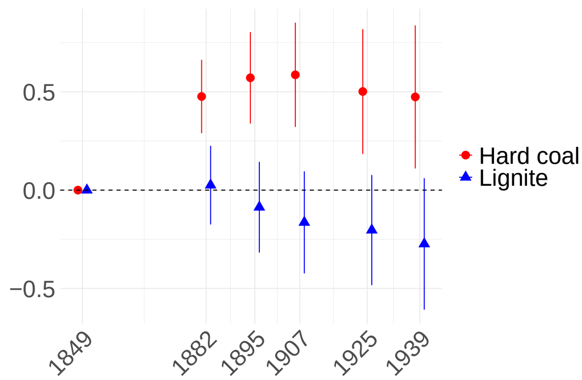
(a) Hard Coal



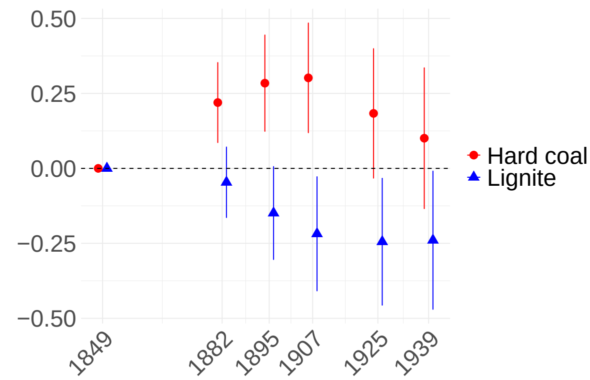
(b) Lignite

Figure A11: Event-Study Estimates: Effect of Coal Phase-Out on Manufacturing Employment (1882–2024, all Post 2007 Years included)

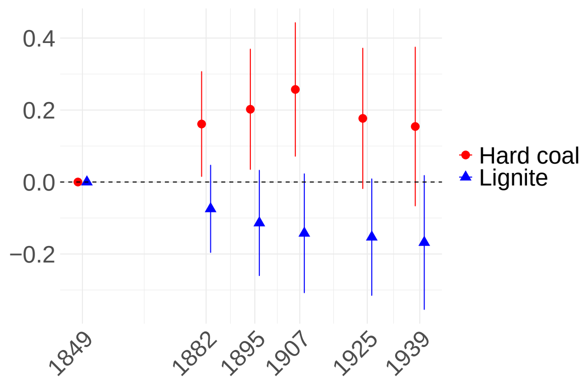
Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each specification interacts either the hard coal (Panel (a)) or lignite (Panel (b)) treatment indicator with year fixed effects. In each case, the other coal region is excluded from the sample. The omitted base year is 1961 for hard coal and 1987 for lignite. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to West German counties.



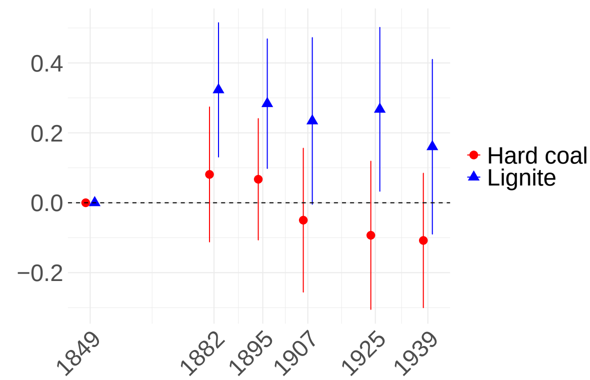
(a) Total employment



(b) Manufacturing employment



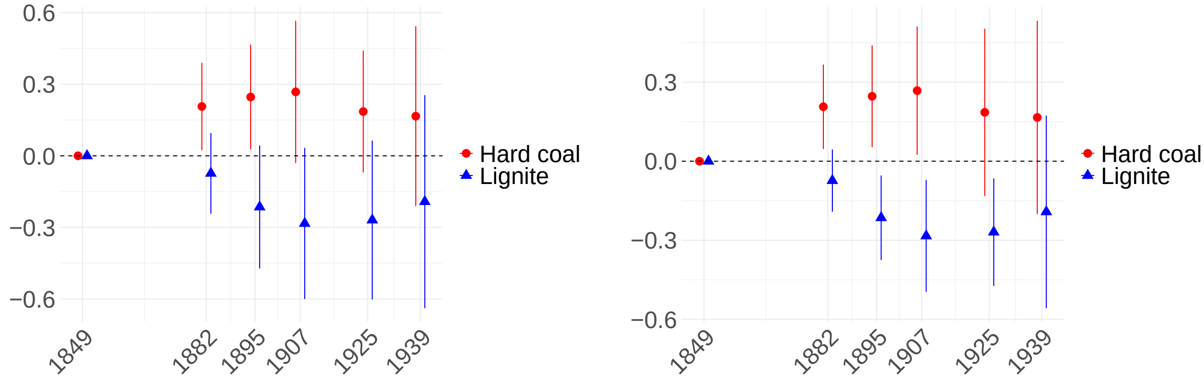
(c) Service employment



(d) Agricultural employment

Figure A12: Event-Study Estimates: Effect of Coal Endowments on Sectoral Employment (1849–1939, Excluding Overlapping Deposits)

Note: This figure plots coefficient estimates and 95% confidence intervals from event-study regressions based on Equation 1. Counties with overlapping deposits of hard coal and lignite are excluded from the analysis. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite indicators are interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.



(a) Conley SE (b) Spatial Inf SE

Figure A13: Event-Study Estimates: Effect of Coal Endowments on Manufacturing Employment (1849–1939, Alternative Inference Methods)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. Panel (a) reports Conley standard errors with a 100 km cut-off (Conley, 1999). Panel (b) implements the spatial inference method proposed by S. O. Becker et al. (2025), which adjusts for spatial unit roots. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The sample is restricted to Prussian counties.

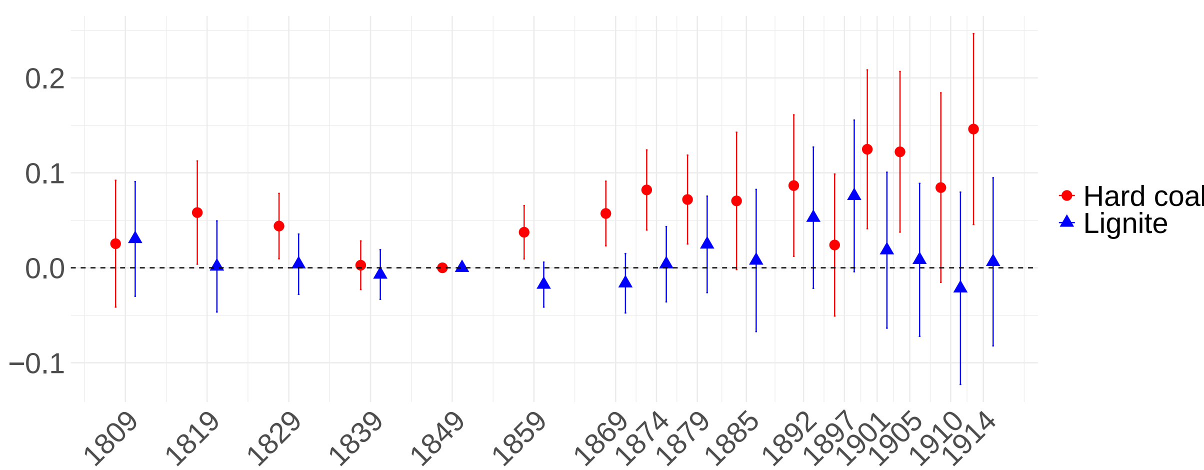


Figure A14: Event-Study Estimates: Effect of Coal Endowments on Wages (1800–1914)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1, using unskilled wages as the outcome variable (see Section A.4 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. The regression includes county and year fixed effects, geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects as well as wheat prices in the nearest market. Standard errors are clustered at the county level. The level of observation is the county, and the sample is restricted to Prussian counties.

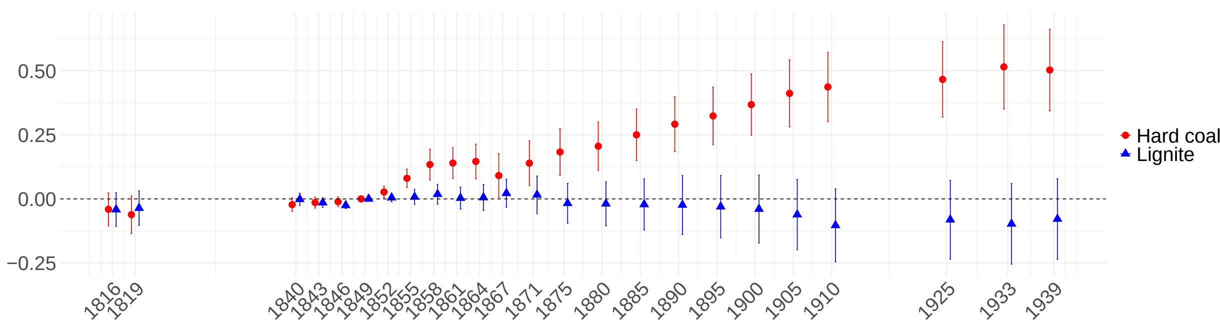
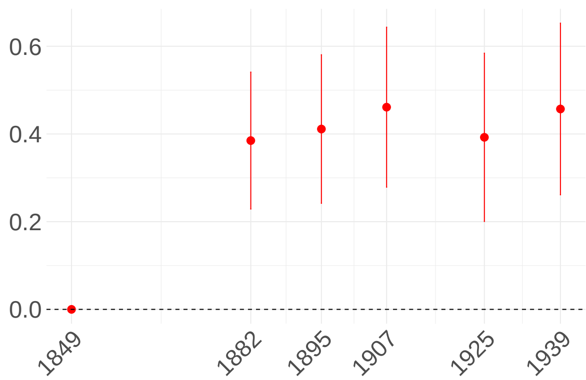
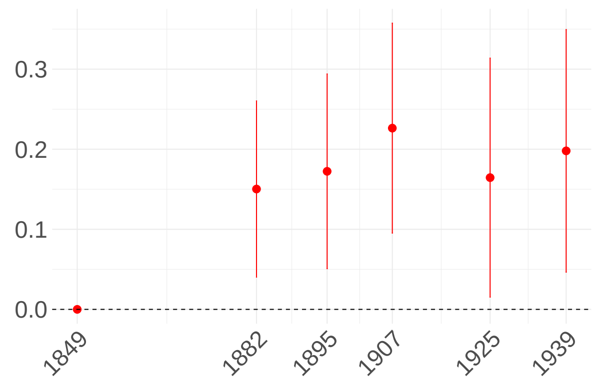


Figure A15: Event-Study Estimates: Effect of Coal Endowments on Urban Population (1816–1939)

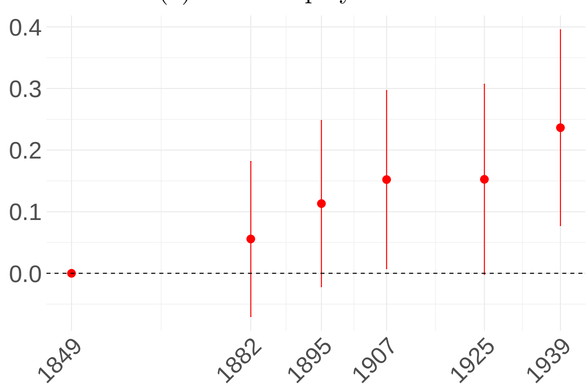
Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1, using urban population (log) as the outcome variable (see Section A.4 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. The regression includes county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the city level. The level of observation is the city, and the sample is restricted to Prussian counties.



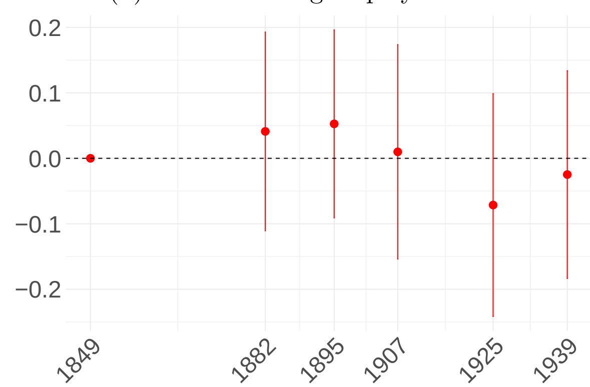
(a) Total employment



(b) Manufacturing employment



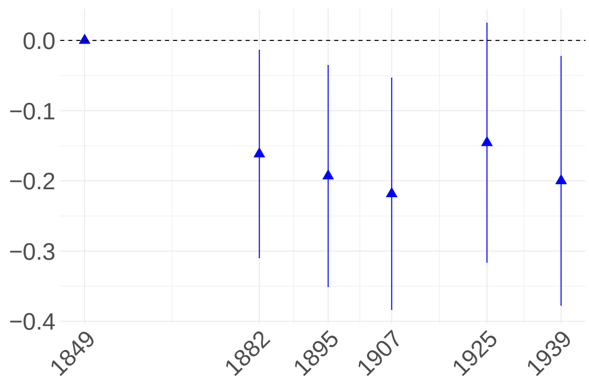
(c) Service employment



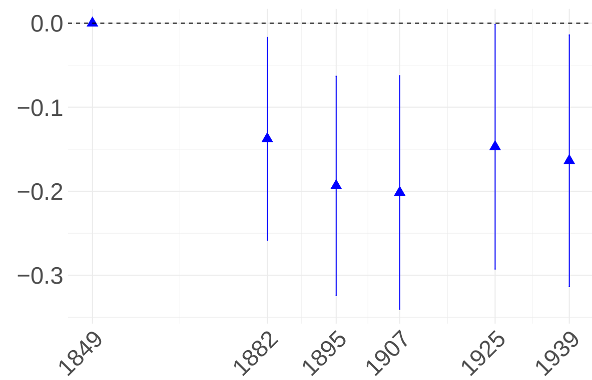
(d) Agricultural employment

Figure A16: Event-Study Estimates: Effect of Hard Coal Endowment on Sectoral Employment (1849–1939)

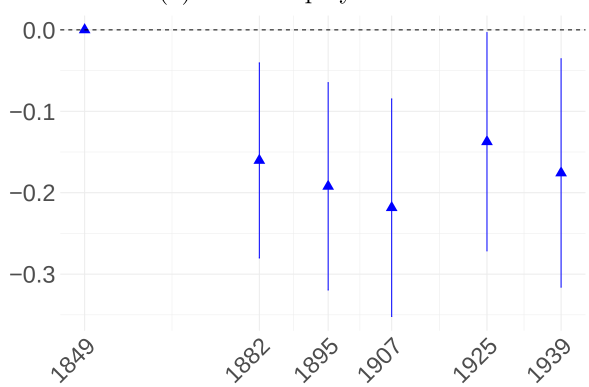
Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.



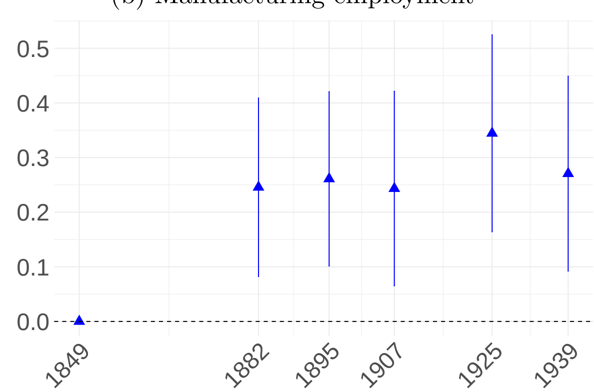
(a) Total employment



(b) Manufacturing employment



(c) Service employment



(d) Agricultural employment

Figure A17: Event-Study Estimates: Effect of Hard Coal Endowment on Sectoral Employment (1849–1939)

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of exposure on county-level employment outcomes: (a) total employment (excluding mining), (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects. The omitted year is 1849. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

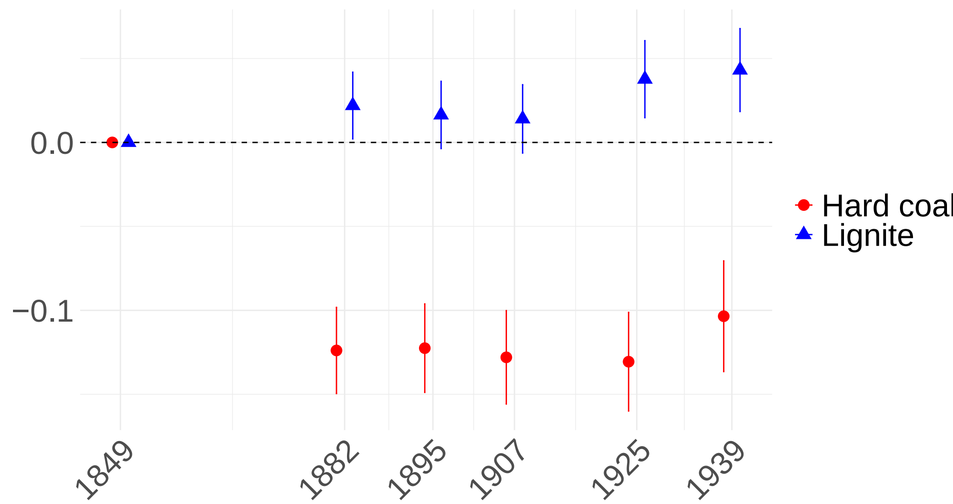
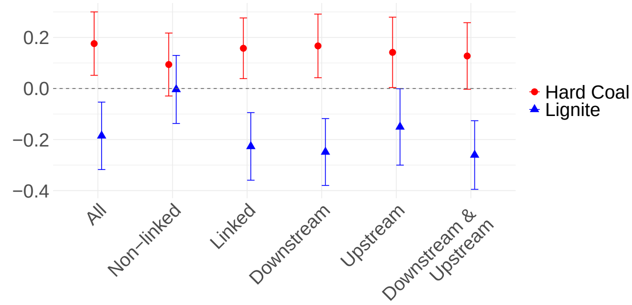


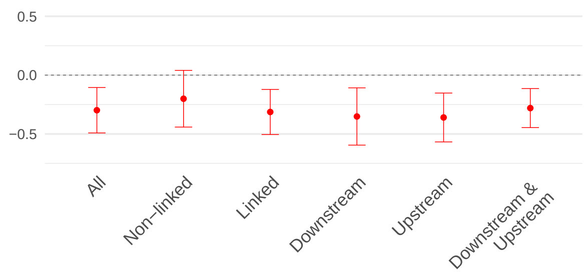
Figure A18: Event-Study Estimates: Effect of Coal Endowment on Sectoral Concentration of Manufacturing Employment (1849–1939).

Note: This figure plots point estimates and 95% confidence intervals from event-study regressions based on Equation 1, using the Herfindahl–Hirschman Index of manufacturing employment as the outcome variable. Higher index values reflect greater concentration, while lower values indicate a more diversified structure (see Section 4.1 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. The outcome is logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects. The omitted year is 1849. The regression includes county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

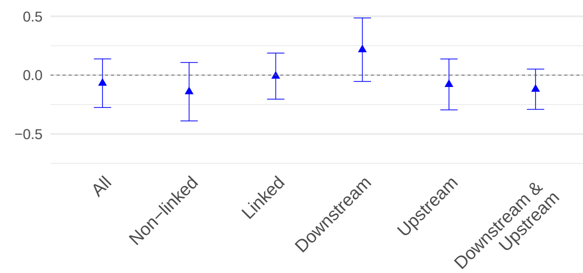
A.2.3 Difference-in-differences Estimates



(a) Pre: 1849; Post: 1882–1939



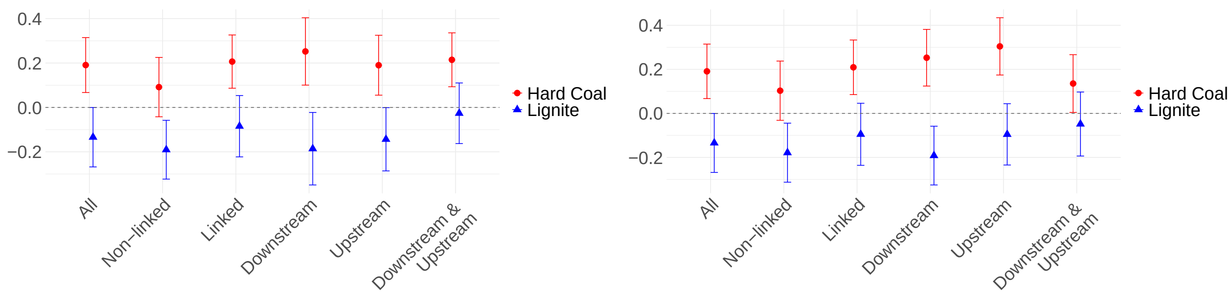
(b) Pre: 1882–1939; Post: 2007–2024



(c) Pre: 1882–1939; Post: 2007–2024

Figure A19: Difference-in-Differences Estimates: Effect of Coal Endowment on Disaggregated Manufacturing Employment using German I-O table

Note: This figure plots point estimates and 95% confidence intervals from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. The outcome variable is manufacturing employment, disaggregated by linkage to coal mining, classified using the German 1936 Input-Output table (Fremdling & Staeglin, 2014). Outcomes are log-transformed and standardized to have a mean of zero and a sd of one. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. Panel (a) shows results for the coal extraction period using counties within Prussian borders. The post-period includes all years from 1882 to 1939, with 1849 as the baseline year. Panels (b) and (c) show results for the coal phase-out period using counties in West Germany. The post-period includes all years from 2007 to 2024, with 1882 to 1939 as the pre-period. Panel (b) reports DiD estimates of the effect of hard coal endowment on manufacturing employment, excluding lignite regions from the sample. Panel (c) reports corresponding estimates for lignite endowment, excluding hard coal regions. Standard errors are clustered at the county level.



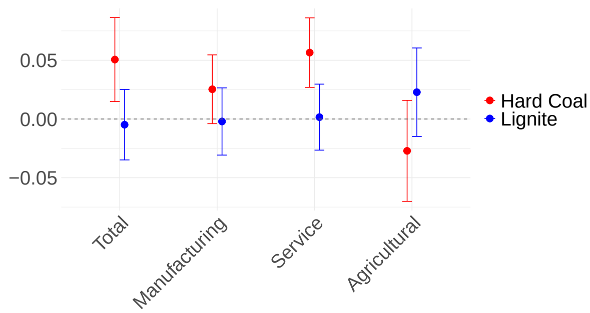
(a) Direct Linkages Only

(b) Indirect Linkages

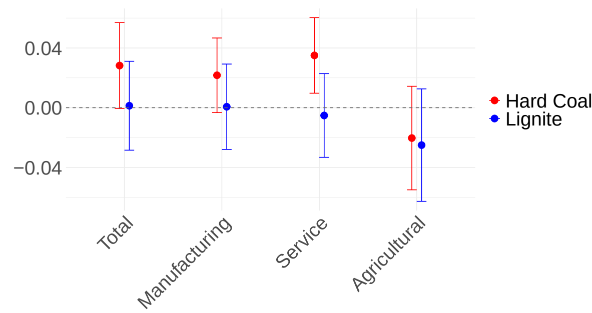
Figure A20: Difference-in-Differences Estimates: Alternative Linkage Definitions

Note: This figure plots point estimates and 95% confidence intervals from difference-in-differences regressions based on Equation 1, with event-time indicators replaced by a single post-period indicator. Both panels report the estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining, based on the U.S. 1947 input–output table (U.S. Bureau of Economic Analysis, 2019). All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with the post-period indicator. The analysis covers the extraction period (1849–1939) and is restricted to counties within Prussian borders. The post-period indicator equals one for all years after 1849. The left panel uses direct linkages only; the right panel uses indirect linkages for both upstream and downstream relationships. Standard errors are clustered at the county level.

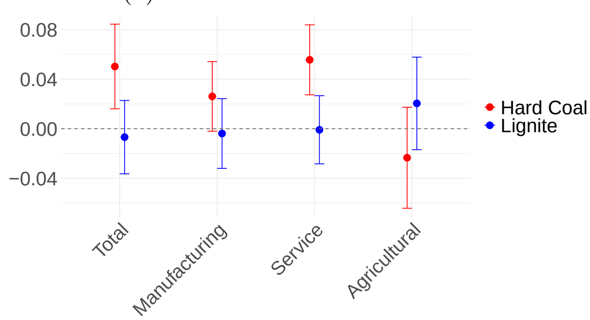
A.2.4 Shift-Share Estimates



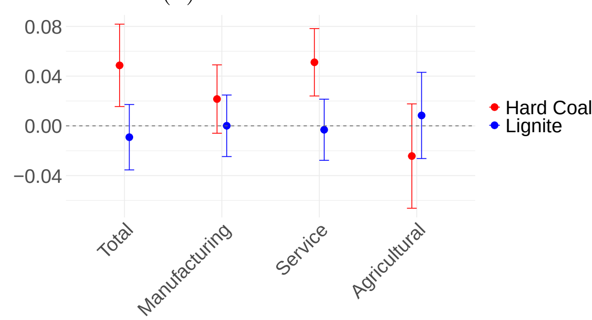
(a) German Coal Production



(b) German Coal Price



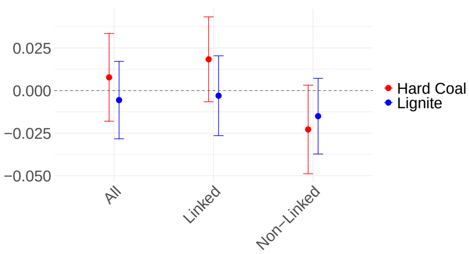
(c) German Coal Revenue



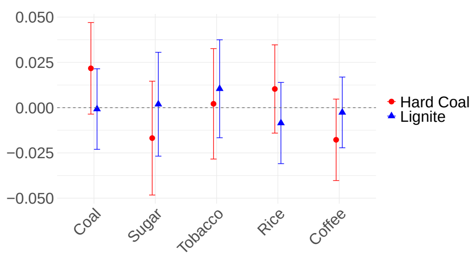
(d) World Coal Price (5 Year Avg.)

Figure A21: Reduced-Form Shift-Share Estimates: Alternative Time-Series Variation

Note: This figure plots point estimates and 95% confidence intervals from reduced-form shift-share approach based on Equation 2, using alternative "shift" variables. Panel (a) uses German hard coal and lignite production; Panel (b) uses prices; Panel (c) uses revenue (price times quantity); and Panel (d) uses a 5-year moving average of global coal prices. Each estimate captures the differential effect of national hard coal (red) and lignite (blue) booms on counties with corresponding coal endowments, relative to non-endowed counties. The dependent variables are county-level employment in total, manufacturing, services, and agriculture. All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.



(a) Disaggregated by Linkage



(b) Placebo Price Shocks

Figure A22: Reduced-Form Shift-Share Estimates: Disaggregated Manufacturing Employment by Linkage

Note: This figure plots point estimates and 95% confidence intervals from reduced-form shift–share regressions based on Equation 2. Each estimate captures the differential effect of global price shocks on manufacturing employment. Panel (a) disaggregates manufacturing employment by linkage intensity to coal mining, showing responses to global coal price shocks in counties with hard coal and lignite endowments, relative to non-endowed counties. Panel (b) presents placebo estimates using global price shocks for unrelated commodities (sugar, tobacco, rice, coffee) to validate the identification strategy. All outcomes are log-transformed and standardized to mean zero and unit variance. All regressions include county and year fixed effects as well as geographic controls (*temperature, rainfall, slope, caloric suitability, distance to rivers, and distance to coast*) interacted with year fixed effects. Standard errors are clustered at the county level. The sample covers the period 1882–1939 within the borders of the German Empire.

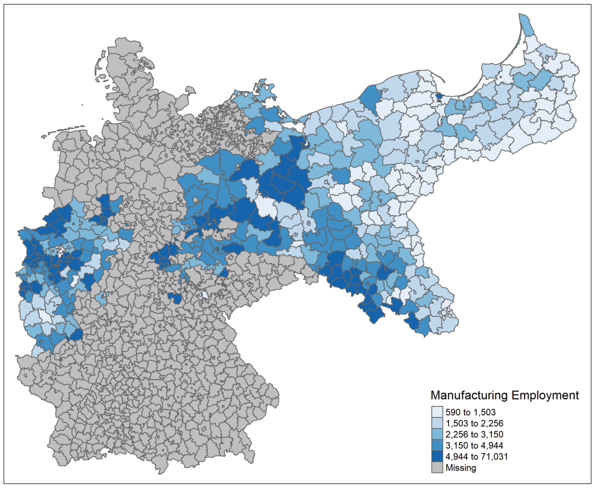
A.3 Spatial Coverage and Harmonization of Administrative Boundaries

Administrative county borders changed repeatedly over the course of my study period due to wars, shifts in sovereignty, and territorial-administrative reforms. To ensure consistency in spatial units across time, I harmonize all data to the geographic boundaries of counties as defined in 1900. Specifically, I overlay historical shapefiles from each census year onto the 1900 county map and aggregate data by weighting historical counties according to the proportion of their area overlapping with 1900 boundaries.

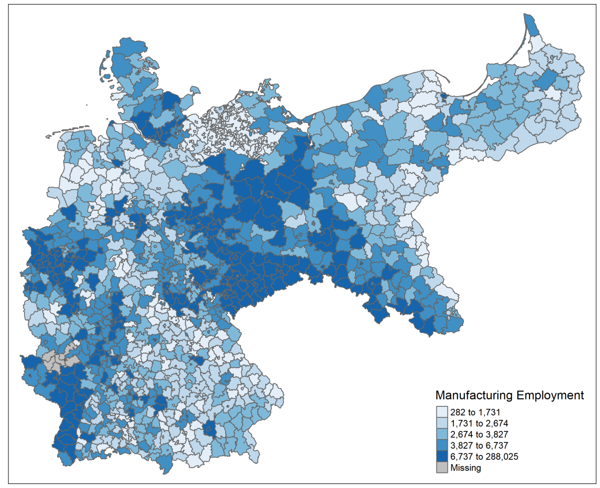
Territorial coverage also varies by year. For instance, the 1849 data include only the Kingdom of Prussia, which at the time encompassed large parts of what would later form the German Empire. In contrast, some eastern regions that were part of Germany in earlier decades, such as Silesia and East Prussia, lie outside present-day borders and are not covered in more recent census waves. To address this, I conduct separate analyses for periods with largely stable territorial coverage and provide robustness checks using balanced panels of counties observed throughout.

Figures A23 and A24 in the Appendix illustrate the geographic coverage of my dataset over time. Blue-shaded counties indicate regions for which manufacturing employment data are available in each year, while grey-shaded areas correspond to regions for which data are missing or which were not part of Germany at the time. These maps highlight how the availability of consistent county-level data varies across historical periods due to administrative boundary changes, wars, and political realignments.

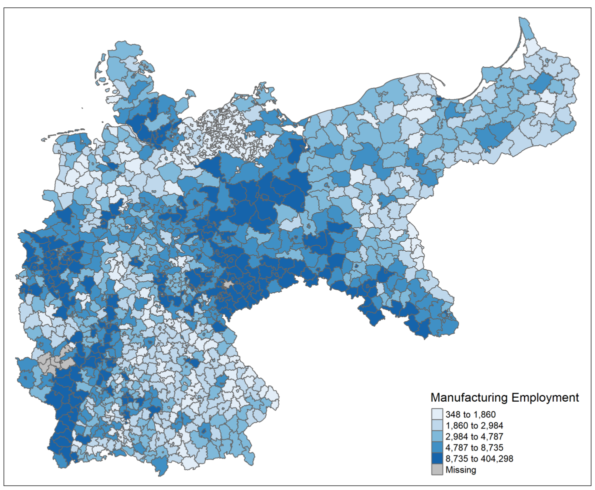
For the 1849–1939 period, I restrict the analysis to counties within Prussia, where the earliest employment data are available ($N = 417$). For the period 1882–1939, the sample includes counties within the borders of the German Empire ($N = 1122$). For the coal phase-out period (1950–2024), I focus on counties in West Germany (Federal Republic of Germany), excluding the former German Democratic Republic (GDR), which offer largely stable and continuous geographic coverage over time ($N = 640$).



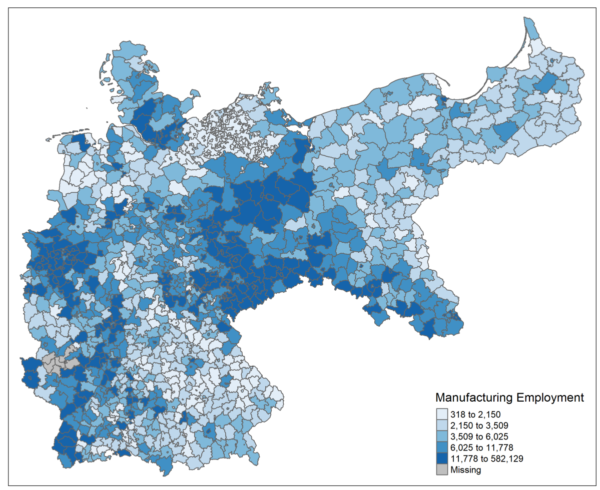
(a) 1849



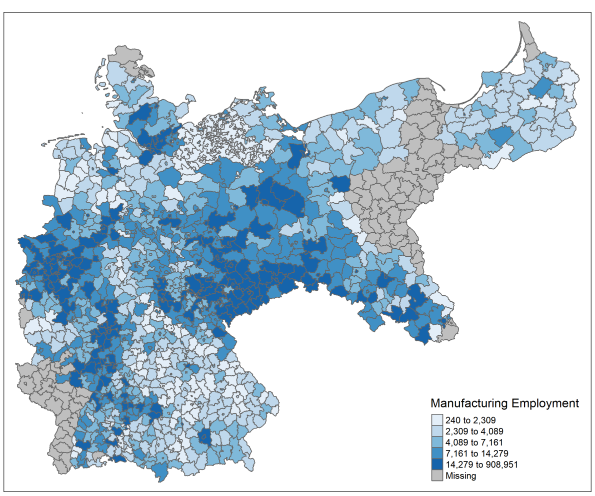
(b) 1882



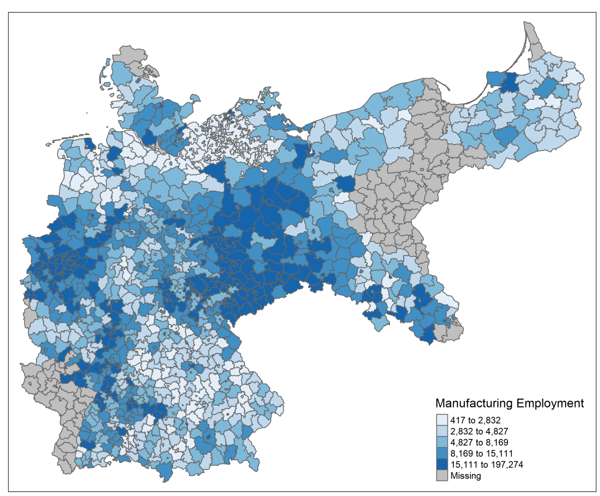
(c) 1895



(d) 1907

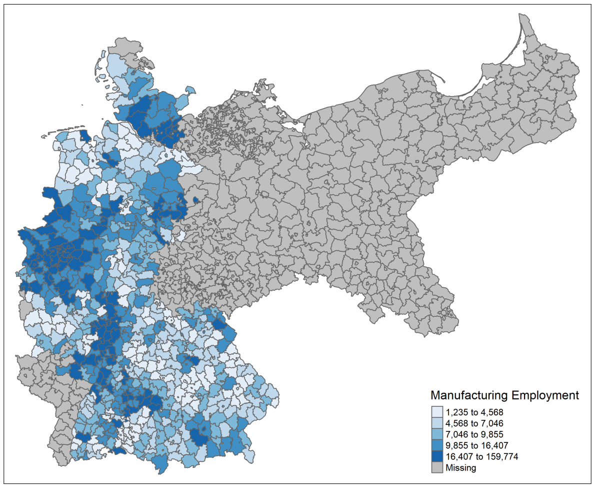


(e) 1925

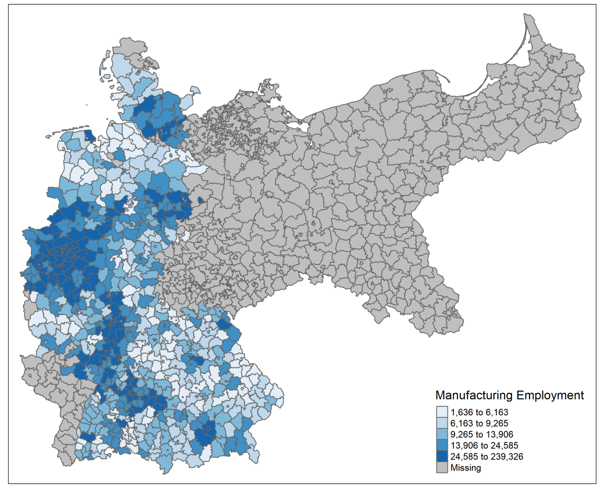


(f) 1939

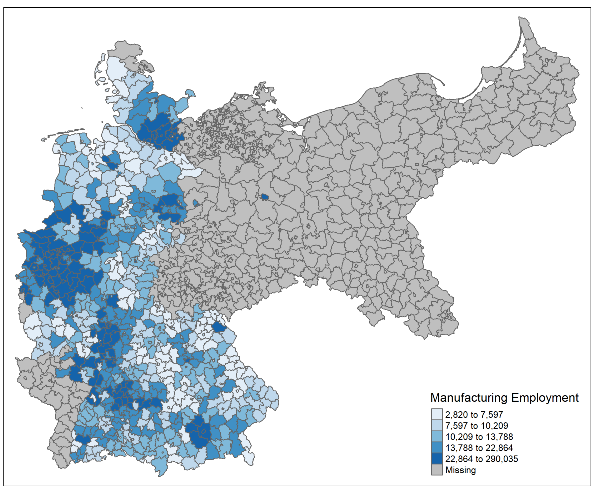
Figure A23: Manufacturing employment in German Counties (1849–1939)



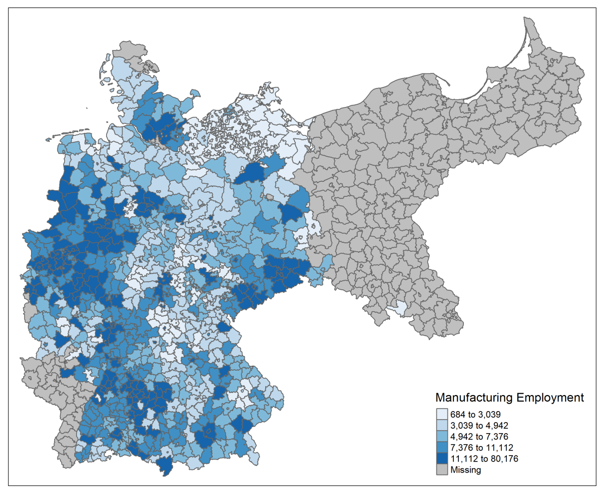
(a) 1950



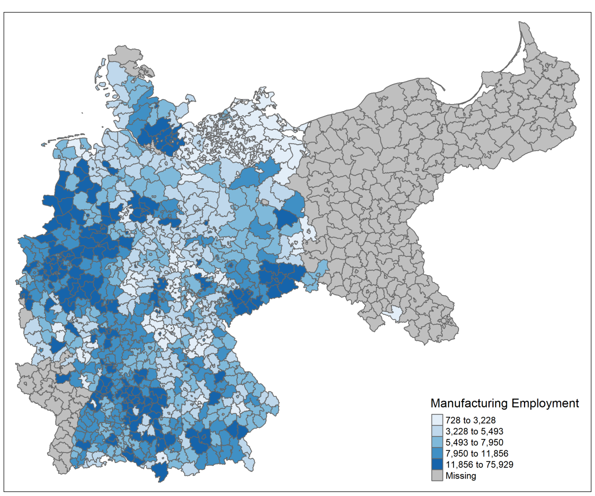
(b) 1961



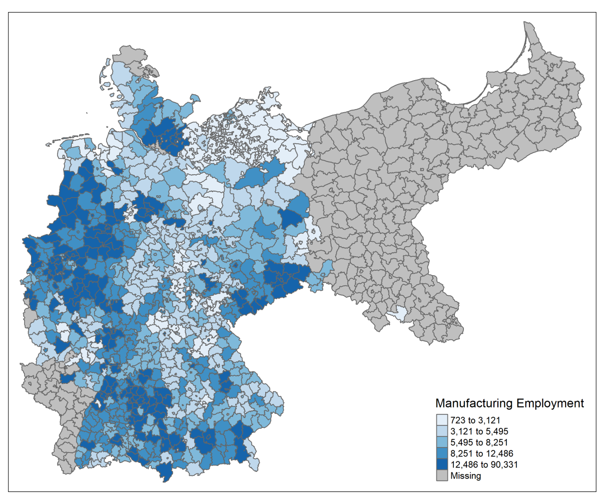
(c) 1987



(d) 2007



(e) 2015



(f) 2024

Figure A24: Manufacturing employment in German Counties (1950–2024)

A.4 List of Variables and Data Sources

A.4.1 Outcome Variables

Sectoral Employment 1849–2024: County-level employment data for 1849 is drawn from Statistisches Bureau zu Berlin (1851–1855). Data for the years 1882, 1895, 1907, 1925, 1933, and 1939 are taken from the respective occupation censuses in Kaiserliches Statistisches Amt, 1884, 1897, 1910; Statistisches Reichsamt, 1929, 1936b, 1943. Employment data for 1950, 1961, 1970, and 1987 are sourced from the Statistisches Bundesamt, 1953, 1964, 1972, 1989. Annual employment data from 2007 to 2024 are obtained from the Bundesagentur für Arbeit (2025).

Unskilled Wages 1800–1914: Average daily wage in Mark of unskilled male "seasonal fill" workers (day laborers) employed in public state forestry during the time periods 1800–1809, 1810–1819, 1820–1829, 1830–1839, 1840–1849, 1850–1859, 1860–1869, 1870–1874, and 1875–1879 as reported by Eggert (1883). Average local daily wages in Mark for manual day laborers in a county for the year 1883 as reported by Schmitz, 1888 and for the years 1892 to 1914 as reported by Zentralblatt, 1892. The complete digitized wage series is compiled by A.-K. Becker and Hornung (2025).

Urban Population 1816–1939: The total number of inhabitants in cities with city rights for 24 years between 1816 and 1939, originally published by the Prussian Statistical Office and compiled by Hornung (2015) and Matzerath (1985). The dataset covers all urban municipalities in Prussia with more than 5,000 inhabitants.

A.4.2 Treatment Variables

Hard Coal (Indicator): Binary variable that takes a value of one for counties with access to hard coal resources within Germany, defined as those with rock strata formed during the Carboniferous period (approximately 295 to 320 million years ago) within 20 km of the county's geodesic centroid; zero otherwise. Data source: Asch and Bellenberg, 2005.

Lignite (Indicator): Binary variable that takes a value of one for counties with access to lignite resources within Germany, defined as those with rock strata formed during the Miocene, Oligocene, Paleogene, or Pliocene period (approximately 2 to 66 million years ago) where lignite is among the two most prevalent rock types within 20 km of the county's geodesic centroid; zero otherwise. Data source: Asch and Bellenberg, 2005.

Hard Coal Field (Continuous): This variable measures the distance in km from a county centroid to the nearest hard coal field. The underlying geospatial data on major nineteenth-century coalfields in Europe is taken from Châtel and Dollfus (1931) and digitized by Fernihough and O'Rourke (2021). Because the dataset includes both lignite and hard coal fields, I manually classified each site based on historical sources of mine locations, retaining only hard coal fields. The distance measure is truncated at zero, that is, if a county centroid lies directly on a hard coal field, the value is set to zero.

Lignite Field (Continuous): This variable measures the distance in km from a county centroid to the nearest lignite field. The underlying geospatial data on major nineteenth-century coalfields in Europe is taken from Châtel and Dollfus (1931) and digitized by Fernihough and O'Rourke (2021). Because the dataset includes both lignite and hard coal fields, I manually classified each site based on historical sources of mine locations, retaining only lignite fields. The distance measure is truncated at zero, that is, if a county centroid lies directly on a lignite coal field, the value is set to zero.

A.4.3 Shift Variables

Global Coal Prices 1882–1939: I use annual real world coal prices from Jacks (2019), expressed in 1900 U.S. dollars, for all years between 1882 and 1939.

Global Prices 1882–1939: I use annual real world prices for rice, tobacco, sugar, and coffee from Jacks (2019), expressed in 1900 U.S. dollars, for 1882–1939.

German Hard Coal and Lignite Production, Prices, and Revenues 1882–1939: Data on German hard-coal and lignite prices and production for 1882–1939 come from (Fischer, 2011). Revenues for each resource are computed as the product of the average national production quantity and the corresponding average national price in a given year.

A.4.4 Geographic Control Variables

Temperature: The average temperature (in degrees Celsius) in a county during the 1970–2000 period, constructed by temporally and spatially aggregating time series data on mean monthly temperature at a geospatial resolution of 30 arc seconds (i.e., grid cells of approximately 1 square kilometer), obtained from the WorldClim (version 2.1) dataset (Fick & Hijmans, 2017).

Rainfall: The average precipitation (in millimeters) in a county during the 1970–2000 period, constructed by temporally and spatially aggregating time series data on total monthly precipitation at a geospatial resolution of 30 arc seconds (i.e., grid cells of approximately 1 square kilometer), obtained from the WorldClim (version 2.1) dataset (Fick & Hijmans, 2017).

Slope: Average land slope (in degrees) within a county, derived from geospatial elevation data at a resolution of 30 arc seconds (approximately 1 km $\times$ 1 km). Slope is calculated using the eight nearest neighboring cells, based on the WorldClim version 2.1 dataset (Fick & Hijmans, 2017). This measure captures local terrain steepness, with higher values indicating steeper gradients.

Distance to Coast: The distance in km from a county centroid to the nearest coast, constructed using a map of all European coastlines provided by the European Environment Agency (EEA) (https://www.eea.europa.eu/data-and-maps/data/eea-coastline-for-analysis-1/gis-data/europe-coastline-shapefile)

Distance to Navigable River: The distance in km from a county centroid to the nearest navigable river, constructed using a map of waterways (Schiffahrtsstraßen) in the German Customs Union (Zollverein) in 1850, available through the "Server for Digital Historical Maps" at the Leibniz Institute of European History at the University of Mainz (IEG, 2010).

Caloric Suitability Index:
The average caloric suitability index in a county, as constructed by Galor and Özak, 2016. This index reflects potential agricultural productivity (measured in calories) in 5'x5' grid cells, based on the crops available for cultivation.